Insurance Inside and Outside the Firm

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Abstract

This paper presents a new model with optimal wage contracts, assets, and search frictions in general equilibrium. Contracts are subject to hidden search and limited commitment on the side of workers and firms, and workers can trade risk-free bonds subject to borrowing constraints. In this model, firms take advantage of workers' access to financial markets when they design how wages change with tenure and after productivity shocks. The optimal contract implies a pass-through of productivity shocks to wages and a marginal propensity to consume that are consistent with the data. The insurance that workers receive outside the firm crowds out the insurance that they receive inside the firm: when workers can trade assets, firms choose to pass productivity shocks through to wages significantly more. Similarly, policies improving the ability of workers to self insure are twice less effective at reducing consumption risk because of the endogenous response of optimal wage contract.

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1 Introduction

What determines the risk that workers face in labor markets and how much insurance do workers receive against it? The existing literature has tackled this issue from two distinct perspectives. The first perspective argues that firms provide some form of insurance to workers through wage contracts (Knight, 1921, Baily, 1974, Azariadis, 1975). In response to productivity shocks, firms can either absorb them into their profits, or pass them on to workers through their wages. Recent evidence shows that the pass-through of productivity shocks to wages is small but positive (Guiso, Pistaferri and Schivardi, 2005), suggesting that firms provide significant but incomplete insurance to workers. The second perspective in the literature argues that workers self-insure against labor income risk by accumulating precautionary savings (Bewley, 1977, İmrohoroğlu, 1989, Huggett, 1993, Aiyagari, 1994). Recent evidence on the marginal propensity to consume (Johnson, Parker and Souleles, 2006, Blundell, Pistaferri and Preston, 2008) suggests that households can smooth transitory income shocks, though imperfectly.

Despite decades of research on this issue, there has been no attempt to bring these two perspectives together. The existing literature on optimal wage contracts between firms and workers (e.g. Thomas and Worrall, 1988, Beaudry and DiNardo, 1991, Balke and Lamadon, 2022) makes the stark assumption that workers have no access to financial markets so that firms are the only source of insurance in the economy. This assumption is at odds with evidence on marginal propensity to consume and potentially overstates the importance of firms as insurance providers. By contrast, the models used in the literature on precautionary savings (e.g. Kaplan and Violante, 2014, Auclert, Rognlie and Straub, 2024) are consistent with evidence on marginal propensity to consume but make the assumption that wage contracts are exogenous. This assumption is at odds with evidence on pass-through and implies that wage contracts do not respond to changes in economic policy. As a result, we know very little about the interaction between the insurance that workers receive *inside the firm*, through wage contracts, and the insurance that workers receive *outside the firm*, through financial markets.

This paper builds a new model with optimal wage contracts, assets, and search frictions in general equilibrium, which is consistent with evidence on pass-through and marginal propensity to consume. The model brings together the dynamic wage contracting model of Menzio and Shi (2010) with the canonical precautionary savings models in the tradition of Bewley (1977). I first characterize the contract and show that firms take advantage of workers' access to financial markets when they design how wages change with tenure and in response to productivity shocks. I then estimate the model using ad-

ministrative data from France and evidence on the marginal propensity to consume and use it to evaluate the determinants of earnings inequality and the sources of insurance that workers receive against shocks. In terms of income inequality, I find that the assets workers have when they starts their career, or the assets of their parents, are an important determinant of the life-cycle earnings of workers. In particular, wealthy workers end up receiving higher wages than poor workers because they select wage contracts that are more backloaded but pay more on average. In terms of insurance against shocks, I find that the insurance that workers receive outside the firm, by trading risk-free bonds, significantly crowds out the insurance that workers receive inside the firm, through wage contracts. Specifically, optimal contracts offset about half of the improved insurance that workers enjoy from trading risk-free bonds. Similarly, policies that improve workers' access to financial markets, such as relaxing borrowing constraints, are significantly less effective at reducing consumption risk because firms respond by providing less insurance to workers. In most of the analysis, I assume that assets are public information to workers and firms but later argue that relaxing this assumption makes little difference.

The model features risk-averse workers, risk-neutral firms and dynamic wage contracts with directed search. Workers transition between employment and unemployment, and can switch jobs. They can trade non-contingent assets subject to a borrowing constraint. Firms face firm-level productivity shocks and choose as part of the contract how much to optimally pass them on to workers. Firms and workers also face exogenous unemployment shocks. Contracts are subject to two sets of frictions. First, the search decision of workers is private information, which leads to moral hazard. Second, workers cannot commit to transfers to previous employers after employer-to-employer (EE) transitions and firms cannot commit to transfers to workers after employment-to-unemployment (EU) transitions. The assumption that firms cannot make transfers to workers after EU transitions is meant to capture that not all risk is insurable by firms, so that assets are used for precautionary savings. In the baseline model, assets are public information so firms and workers can contract on the saving decision of workers and firms know the assets of the workers they match with.

The optimal contract satisfies a system of two optimality conditions. The first is a standard consumption growth condition, which also arises in optimal wage contracts with hand-to-mouth workers (e.g. Balke and Lamadon, 2022). The second is a new pseudo Euler equation, which highlights the different roles that assets play in optimal wage contracts. To understand how the presence of assets influences the optimal contract, consider first how this pseudo Euler equation pins down the path of consumption and assets given a path of wages. On one hand, workers whose wages are increasing over time want to

consume their existing assets in order to smooth consumption. On the other hand, these workers also want to accumulate precautionary savings to smooth their consumption after income shocks, for example due to an unemployment shock. Thus, the optimal contract uses the workers' assets to balance their desire to smooth consumption over time with their desire to smooth consumption across states. The pseudo Euler equation also shows that, in addition to being used to smooth the worker consumption, assets are manipulated by firms to influence the search decision of workers through wealth effects as in Acemoglu and Shimer (1999). However, I show in the quantitative model that this effect is minuscule and thus does not influence optimal contracts much.

Consider now how the presence of assets changes the optimal degree of wage backloading in the wage contract offered by firms, and thus the average wage growth. In designing contracts, firms face a trade-off between retaining and attracting workers. Remember that firms cannot control the worker search decision because it is private information. As a result, the optimal retention strategy for firms is to backload wages to make workers search for jobs with higher wages and lower job finding rates as in Burdett and Coles (2003) and Shi (2009). Firms know that in response to backloaded wages, workers will smooth their consumption over time by consuming their existing assets. Therefore, relative to a model without assets firms can potentially backload wages significantly more. However, firms also know that workers face borrowing constraints so they might not be able to smooth consumption over time. Besides, they know that workers would like to save to self-insure against unemployment risk. In particular, backloading wages too much and making workers consume their savings at the beginning of jobs makes contracts very unattractive to risk-averse workers who anticipate that their consumption will fall significantly if an unemployment shock occurs. Indeed, if one firm adopted a strategy of extreme wage backloading, it would have to offer much higher average wages to make its offer attractive relative to an offer that offers more stable wages and, hence, more stable consumption. The optimal contract thus balances the desire of firms to retain workers through backloaded wages, and the desire of workers to smooth consumption and self-insure against unemployment risk through frontloaded wages.

Next, consider how the presence of assets changes the optimal degree of pass-through in wage contracts. Here again, the optimal pass-through of productivity shocks to wages is the result of a trade-off between retaining and attracting workers. To attract workers, it is optimal for firms to maintain consumption independent of productivity with a low pass-through. However, with such a strategy workers would leave at a constant rate. The firm can increase its profits by raising the wage of workers when productivity is high to reduce the quit rate, and lowering the wage when productivity is low to increase it. With

this strategy firms would retain workers precisely when they generate the most profits. When workers can access financial markets, they value stable wages relatively less because they can smooth consumption themselves. As a result, firms choose to provide less insurance to workers against productivity shocks and instead optimize worker retention more. Thus, letting workers trade assets increases the pass-through of productivity shocks to wages.

Allowing workers to trade assets influences optimal contracts, even when those trades are public information, because of the limited commitment of firms and workers. Specifically, I show that if firms and workers can both commit to transfers post EE and EU transitions, then assets do not influence contracts. Besides, if only firms could commit to transfers but not workers, then assets are a substitute for transfers post EE transitions. Conversely, if only workers could commit to transfers but not firms, then assets are a substitute for transfers post EU transitions. In both cases, substituting between assets and either firm or worker commitment does not change the path for consumption or EE probability (in the absence of borrowing constraint), but it changes how these paths are implemented. With limited worker commitment, the optimal contract features extreme wage backloading in that wages are negative upon hiring and assets remain negative throughout the match. With limited firm commitment, the optimal contract features extreme wage frontloading in that workers receive a large upfront wage upon hiring and assets remain positive throughout the match. These results clarify why assets play a role in my model whereas they do not in the literature on optimal unemployment insurance (e.g. Hopenhayn and Nicolini, 1997). In this literature, it is standard to assume that the government (the principal) can tax the worker (the agent) when she finds a job. This assumption is equivalent to assuming worker commitment in my setting[1]. Besides, these results show that elements from the literature on optimal wage contracts and precautionary savings are both critical to the model. In particular, tenure profiles for wages are consistent with the data only with limited worker commitment (as in standard wage contracts) and with limited firm commitment (as in standard precautionary saving models).

I calibrate the model using French administrative data to quantify how firms balance the motive to optimize worker retention with the motive to insure workers against shocks. I estimate model parameters using moments that are informative about the risk that workers face from unemployment on from productivity shocks, and the extend to which firms are competing to retain their workers. In particular, I target worker transitions across employment and unemployment (UE, EE and EU rates), the tenure profile of wages and EE transitions, the standard deviation and persistence of firm productivity shocks, and

[1]Hence the focus of this literature on hidden saving.

an estimate of the marginal propensity to consume out of unemployment shocks. The main empirical success of the model is to generate a marginal propensity to consume out of wage shocks and a pass-through of productivity shocks to wages that are both far smaller than one, consistent with the data. By contrast, existing wage contracting models can match estimates of the pass-through but generate a marginal propensity to consume of one. Conversely, existing precautionary savings models can match estimates of the marginal propensity to consume but generate a pass-through of one. Matching these estimates is critical for my analysis because the fact that they are small implies that workers receive a significant amount of insurance both inside and outside the firm.

I first use the quantitative model to study the determinants of income inequality. The model implies that workers who start their career with no initial assets receive lower average wages than workers who start their career with positive assets. The reason is that poor workers choose to search in markets where wage contracts are less backloaded because they are not able to smooth consumption themselves. However, these contracts also offer lower average wages because firms anticipate the high mobility rate of workers. This mechanism provides a new explanation for the widely documented persistence of income inequality across generations (e.g. Solon, 1992). It is also consistent with evidence that workers with better access to financial markets receive wages that are more backlaoded (Guiso, Pistaferri and Schivardi, 2012) and that workers with wealthy parents experience higher wages growth during their career than workers with poor parents (Halvorsen, Ozkan and Salgado, 2022).

I then use the quantitative model to study the sources of insurance that workers receive against shocks. I find that the insurance that workers receive outside the firm crowds out significantly the insurance that they receive inside the firm. Quantitatively, the pass-through of productivity shocks to consumption is about 2 times lower in my model relative to a model with optimal contract and hand-to-mouth workers. Thus, workers receive more insurance overall. However, the pass-through to wages is also 3 times higher in my model so workers receive less insurance from firms. Thus, endogenous contracts offset some of the benefits of self-insurance in terms of consumption volatility. Because firms provide less insurance to workers, they are able to optimize retention better. In fact, after a positive productivity shocks EE mobility falls more in my model with assets relative to a model with hand-to-mouth workers, and as a result allocative efficiency is higher.

Perhaps the most surprising implication of the model is that firms also take advantage of workers' access to financial markets to influence the shape of the pass-through, and not just its size. To understand how, consider for instance a persistent and positive shock

to firm productivity. Firms respond by cutting the wage of workers on impact, and increasing it significantly later on. Meanwhile, consumption increases smoothly over time as workers initially use their assets in anticipation of higher future income. Eventually, the wage of workers increases so much that assets rise too. Why is it optimal for firms to cut wages and deplete the worker assets on impact after a positive productivity shock? When productivity increases, the profits of firms rise. As a result, the worker retention motive described before becomes stronger and firms choose to backload wages more. They backload wages so much that wages actually fall mildly on impact.

Finally, the model has new implications for the effects of a policy that improves the ability of workers to self-insure by relaxing their borrowing constraint. When workers can borrow, their marginal propensity to consume falls as they are better able to self insure. However, the pass-through of productivity shocks to wages also increases. Overall, I find that endogenous wage contracts offset about half of the insurance gains for workers. Similarly, letting workers borrow allows them to smooth consumption better after an unemployment shock. However, wages are also more backlaoded when workers can borrow so it is more difficult to accumulate precautionary savings. Here again, the endogenous response of optimal contract offset about half of the gains of the policy in terms of reduced consumption risk for workers.

Throughout, I have assumed that assets are public information. This implies that firms know how much assets workers have when they first match, and that the saving decision of workers is contractible. It is natural to wonder whether this assumption, which is somewhat unrealistic, has important implications for the optimal contract. I show that with CARA utility, the allocation is identical when assets are private or public information. This result implies that we can reinterpret the model as one where firms design wage contracts that workers select depending on their assets, and where workers choose how much to consume and save independently of firms. With CRRA utility however, the optimal contract designed under the assumption that assets are public information is no longer incentive compatible because of wealth effects on search. These deviations however appear to be quantitatively small, suggesting that the optimal contract would not be very different if they were taking into account these deviations. Those results are closely related, thought not identical, to those of Werning (2002) and Abraham and Pavoni (2008) on optimal unemployment insurance, and of Chaumont and Shi (2022) and Eeckhout and Sepahsalari (2023) on fixed-wage contracts with assets and directed search.

In this paper I focus on non-contingent assets, as opposed to more general securities, because in the data most assets held by households, such as cash, are non-contingent. This assumption has two important implications relative to the model where workers

have access to complete financial markets studied by Stevens (2004). First, with complete markets the optimal contract no longer suffers from moral hazard. In particular, the worker buys the job from the firm by paying an upfront fee, and then receives a wage equal to the value of output. By contrast, with non-contingent assets, such a contract is not optimal because firms want to insure workers against the risk of not finding another job. As a result, the optimal contract is still subject to moral hazard[2]. Second, the model with complete markets makes predictions that are sharply at odds with the data. In particular, in my model the tenure profile of wages is consistent with the data precisely because workers face uninsurable unemployment risk, and there is limited pass-through of productivity shocks to wages, as in the data, only with incomplete markets.

This paper contributes to a recent literature bringing together labor market transitions and asset accumulation. An early example is Krusell, Mukoyama and Şahin (2010) who study precautionary savings in a DMP model where wages are set by Nash bargaining. Several articles have since introduced assets in search models with EE transitions (Lise, 2013, Chaumont and Shi, 2022, Alves, 2022, Kaas, Lalé and Nawid, 2023, Caratelli, 2024) but in these models wage contracts are subject to ad hoc restrictions. Specifically, wages are assumed to be constant during matches, or assumed to change only when workers receive an outside offer. Instead, my paper is the first to study the determinants of worker mobility and assets in a model with optimal wage contracts. The main advantage of my approach is that it allows to study the sources of insurance that workers receive against labor market risk. Besides, I also show that fixed-wage contracts are inefficient in my environment because they abstract from the firm's desire to retain workers and the worker's demand for precautionary savings. Quantitatively, I find the gains from optimal contracts relative to fixed-wage contracts for firms to be quite large.

The paper starts in section 2 by presenting a new model with wage contracts and assets, which I characterize in section 3. Section 4 brings the model to data and quantifies determinants of income inequality and the amount of insurance that workers receive inside and outside the firm. Finally, section 5 revisits the assumption that assets are public information. Proofs are in the appendix.

2 A model with wage contracts and assets

I first present a new model with search frictions, dynamic wage contracts and assets. The model combines the optimal contract with search frictions of Menzio and Shi (2010) and

[2]This is similar to what Shimer and Werning (2008) call the need to "insure against uncertain spell duration" in the unemployment insurance literature.

the precautionary savings model of Bewley (1977).

2.1 Environment

Time is discrete and runs forever.

Agents A continuum of ex-ante homogeneous workers can be employed or unemployed. Workers receive wage w when employed, and home production b when unemployed. They have period utility $u(c)$ over consumption and discount the future at rate β.

Firm are owned by foreign diversified investors, so they are effectively risk-neutral with discount rate r. An active firm is one that is matched with a single worker. The output from that match is x_t with firm productivity x_t following the mean reverting process

$$x_t = (1-\rho)\overline{x} + \rho_x x_{t-1} + \sigma_x \nu_t$$

where ν_t are i.i.d. innovations with standard normal distribution, and ρ_x parametrizes the persistence of productivity. Firm fixed productivity $\overline{x}$ is drawn at the start of the match, independently across firms. This productivity stays constant over time and lasts for the length of the match.

Financial markets This is a small open economy with foreign interest rate r. Workers can save using risk-free bonds a_{t+1} subject to a borrowing constraint, so that

$$a_{t+1} \geq 0$$

Timing Each period, the sequence of events is as follows

a) Firm productivity shocks and exogenous separations into unemployment occur

b) Employed and unemployed workers (at the start of the period) search for jobs; firms post vacancies; new matches are formed and new contracts are signed

c) Firms produce and pay current wages; workers make saving decision and consume

Directed search with assets There is a continuum of labor markets indexed by the promised value to a worker denoted v and the assets of workers a. I assume that firms commit to never match with workers who have a different level of assets $\tilde{a} \neq a$. Since asset are public information, this assumption ensures that workers will only search in the

labor market that was designed for them, and therefore that firms know the asset of the workers they match with when they design wage contracts. In section 5, I relax the assumption that assets are public information and evaluate whether, as a result, a worker with asset a would benefit by searching in markets indexed by $\tilde{a} \neq a$.

Every period, workers with asset a choose in which labor market v to search, and firms choose where to post vacancies. Both employed and unemployed workers search in the same labor markets. Firms post vacancies in these markets and, only after they match, learn about their productivity x and $\overline{x}$.

Denote $\phi_u(v,a)$ and $\phi_e(v,a)$ the mass of unemployed and employed workers searching for a job and denote $\phi_f(v,a)$ the mass of vacancies posted by firms. Let κ denote the search intensity of employed workers relative to unemployed workers. In each labor market, a constant returns to scale matching function $\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f)$ turns workers searching for a job and vacancies into matches. Define the job finding rate $\tilde{\lambda}_w(\phi_u + \kappa\phi_e, \phi_f)$ as the probability that an unemployed worker finds a job, and the vacancy filling rate $\tilde{\lambda}_f(\phi_u + \kappa\phi_e, \phi_f)$ as the probability that a vacancy finds a worker. These probabilities are defined in the usual way as

$$\tilde{\lambda}_w(\phi_u + \kappa\phi_e, \phi_f) \equiv \frac{\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f)}{\phi_u + \kappa\phi_e}, \quad \tilde{\lambda}_f(\phi_u + \kappa\phi_e, \phi_f) \equiv \frac{\mathcal{M}(\phi_u + \kappa\phi_e, \phi_f)}{\phi_f}$$

Since these matching probabilities will depend on v and a in equilibrium, we can write them in short-hand notation as

$$\lambda_w(v,a) \equiv \tilde{\lambda}_w(\phi_u(v,a) + \kappa\phi_e(v,a), \phi_f(v,a)), \quad \lambda_f(v,a) \equiv \tilde{\lambda}_f(\phi_u(v,a) + \kappa\phi_e(v,a), \phi_f(v,a))$$

In equilibrium there will be an upper bound $\overline{v}$ on the set of active labor markets. For $v > \overline{v}$, the job finding rate is not defined because no firm post vacancies there. I extend this function by setting it to 0 for these values above $\overline{v}$. This implies that a worker can always choose not to search by selecting $v > \overline{v}$.

Unemployed workers Unemployed workers face a standard consumption-savings decision problem similar to Chaumont and Shi (2022) and Eeckhout and Sepahsalari (2023). They receive unemployment benefits b and choose how much to save and consume. They also choose in which labor market v to search. Given the job finding probability, $\lambda_w(v,a)$,

the value of unemployed workers satisfies

$$\begin{aligned} U(a_t) = \max_{c_t, a_{t+1}, v_{t+1}} \quad & u(c_t) + \beta \left[\lambda_w(v_{t+1}, a_{t+1}) v_{t+1} + (1 - \lambda_w(v_{t+1}, a_{t+1})) U(a_{t+1}) \right] \\ \text{s.t.} \quad & c_t \leq (1+r)a_t + b - a_{t+1} \\ & a_{t+1} \geq 0 \end{aligned}$$

In choosing in which labor market v to search, workers face the following standard trade-off: searching in a high-v labor market brings a higher value v conditional on a match, but it will turn out that these matches occur with lower probability because $\lambda_w(v, a)$ will decrease with the value v in equilibrium.

Employed workers Employer workers also search for jobs. With probability $\kappa \lambda_w(v, a_t)$, they find a new match in market v when their current asset is a_t. Existing matches break up and workers separate into unemployment with exogenous probability δ.

Contracts When firms and workers are first matched, they sign wage contracts which specify wages w_t and transfers conditional on separations into unemployment $(1+r)\tau_t^{eu}$ or employment $(1+r)\tau_t^{ee}$ for each history of shocks and conditional on the worker's initial asset position. Productivity shocks $(\overline{x}, x_t)$ are public information and I assume that it is unfeasible for firms to make counteroffers to their workers when they receive outside job offers[3]. In the baseline model, I assume that the savings decision of workers a_{t+1} is public information so it is contractible but I relax this assumption in section 5.

The contract is subject to two sets of contracting frictions, which both play a critical role in the analysis. First, the worker search decision is private information, which leads to moral hazard. Second, workers and firms have limited commitment in that they cannot commit to transfers post EU or EE separations (τ_t^{eu}, τ_t^{ee}). This implies that these transfers must be 0 in the optimal contract. I show in section 3 that the limited commitment of firms and workers is precisely the reason why assets play a role in the optimal contract.

The assumption that workers cannot commit to transfers post EE separations is natural as bonds between firms and workers are prohibited by law. The assumption that firms cannot commit to transfers post EU separations is more controversial because firms do often make severance payments in practice. However, in the data the consumption of workers drops when they become unemployed, which shows that this is a source of risk that firms have not entirely insured away. Besides, if exogenous separations occur

[3]This assumption can be formally justified as follows: counteroffers are private information to workers, and expire before workers can return to their current employers to negotiation for higher wages. These assumptions ensure that it is optimal for current employers not to respond to outside offers.

because of bankruptcy, then firms are not able to make severance payments to workers. What is critical for the analysis is not necessarily that the absence of severance payment comes from limited firm commitment, but instead that workers face some risk that firms cannot insure entirely and therefore require insurance outside firms in the form of precautionary savings[4].

2.2 Optimal contracts

Following previous work on dynamic contracts, I write the contract recursively in terms of promised values and continuation values instead of histories of shocks[5]. Denote V_t the promised value of a worker at the start of the period. The state of a match at the beginning of the period are the worker promised value V_t, the assets of the worker a_t as well as the current productivity x_t.

The components of the contract at time t are the wage paid today, the transfers post separations, the savings decision of workers and a set of continuation values for each state tomorrow. Formally, these components are represented by the functions

$$w_t\left(V_t, a_t, x_t\right), \quad \tau_t^{eu}\left(V_t, a_t, x_t\right), \quad \tau_t^{ee}\left(V_t, a_t, x_t\right), \quad a_{t+1}\left(V_t, a_t, x_t\right), \quad V_{t+1}(V_t, a_t, x_t, x_{t+1})$$

A *contract* is a collection of these functions for all t. In the recursive formulation below, we write the components of the contract as $w_t, \tau_t^{eu}, \tau_t^{ee}, a_{t+1}, V_{t+1}(x_{t+1})$. It will be convenient to denote the continuation value of workers at the current job by

$$W_t \equiv u(c_t) + \beta \mathbb{E}_{x_{t+1}}\left[V_{t+1}(x_{t+1})|x_t\right] \tag{1}$$

This value is pinned down by the contract.

Worker value Given a contract, the worker chooses a search strategy to maximize the present value of utility. The value of a worker satisfies

$$V_t = \delta U(a_t + \tau_t^{eu}) + (1-\delta) \max_{v_t} \left[\kappa\lambda_w(v_t, a_t - \tau_t^{ee})v_t + (1 - \kappa\lambda_w(v_t, a_t - \tau_t^{ee}))\, W_t\right] \tag{2}$$

In equation (2), the first term is the continuation value of a worker who becomes un-

[4]There are other ways to generate such need for insurance outside the firm. For instance, firms can be subject to walk-away constraints specifying that their continuation value must remain positive, which implies that the wage of workers will fall after large negative shocks (as in Thomas and Worrall, 1988). Similarly, if workers receive income shocks that are private information to firms (e.g. health shocks), then the amount of insurance that firms can provide is limited.

[5]I abstract from randomized contracts to keep notations simple.

employed at time t with assets a_t and receive transfer from firms τ_t^{eu}. The second term depends on the probability that a worker finds another job $\kappa\lambda_w(v_t, a_t - \tau_t^{ee})$ and on the value that the worker receives if an EE transition occurs v_t. A worker who finds anew job and starts the period with asset a_t will start her new job with asset $a_t - \tau_t^{ee}$. A worker who does not find a new job receives the continuation value W_t.

Equation (2) shows how the assumptions of hidden search and limited worker commitment interact. First, moral hazard arises because the worker's search decision only depends on the surplus that workers get from EE transitions $v_t - W_t$, and not on the firm value. Thus, relative to the search policy that the worker chooses, the firm with a positive value would prefer the worker to search instead in markets with a higher value v_t and a lower job finding rate $\lambda_w(v_t, a_t - \tau_t^{ee})$ because it wants to retain the worker. Since the worker search decision v_t is private information, the firm cannot control it directly and instead influences the worker's decision by manipulating continuation values. The assumption of limited worker commitment implies that firms can only manipulate the continuation value of workers at the current job W_t and not the probability that workers find another job $\lambda_w(v_t, a_t - \tau_t^{ee})$.

When workers have access to assets a_t, the firm has an additional tool to influence the worker's search decision since assets a_t influence the job finding rate $\lambda_w(v_t, a_t - \tau_t^{ee})$ of workers. Assets however also influence the continuation value of workers who become unemployed $U(a_t + \tau_t^{eu})$. This dual role for assets generates a trade-off that I describe in details in section 3.

Optimal contracts Denote the search policy $v(W_t, a_t - \tau_t^{ee})$, the EE probability as

$$p(W_t, a_t - \tau_t^{ee}) \equiv \kappa\lambda_w(v(W_t, a_t - \tau_t^{ee}), a_t - \tau_t^{ee})$$

and the worker expected surplus from EE transitions as

$$S(W_t, a_t - \tau_t^{ee}) \equiv \kappa\lambda_w(v(W_t, a_t - \tau_t^{ee}), a_t - \tau_t^{ee})\,(v(W_t, a_t - \tau_t^{ee}) - W_t)$$

Finally, denote $\Pi(V_t, a_t, x_t)$ the present value of profits for a firm matched with a worker with promised value V_t and assets a_t when productivity is currently x_t and denote the vector of choice variables $\chi_t \equiv \{w_t, \tau_t^{eu}, \tau_t^{ee} c_t, a_{t+1}, V(x_{t+1})\}$. Taking as given the value of unemployment $U(a)$, the job finding rate $\lambda_w(v, a)$, and the search policy of

workers $v(W,a)$, the optimal contract solves

$$\Pi(V_t,a_t,x_t) = \max_{\chi_t}(1-\delta)(1-p(W_t,a_t-\tau_t^{ee}))\left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}[\Pi(V(x_{t+1}),a_{t+1},x_{t+1})|x_t]}{1+r}\right)$$
$$-\delta(1+r)\tau_t^{eu} + (1-\delta)p(W_t,a_t-\tau_t^{ee})(1+r)\tau_t^{ee}$$

subject to

$$\begin{aligned}
&\text{(PK)}: && V_t \leq \delta U(a_t+\tau_t^{eu}) + (1-\delta)\left[W_t + S(W_t,a_t-\tau_t^{ee})\right]\\
&\text{(Budget)}: && c_t + a_{t+1} = (1+r)a_t + w_t\\
&\text{(BC)}: && a_{t+1} \geq 0\\
&\text{(LC)}: && \tau_t^{eu}, \tau_t^{ee} = 0
\end{aligned}$$

where $W_t = u(c_t) + \beta\mathbb{E}_{x_{t+1}}\left[V_{t+1}(x_{t+1})|x_t\right]$.

The firm maximizes the present value of profits, where $(1-\delta)(1-p(W_t,a_t-\tau_t^{ee}))$ is the probability that the worker remains within the current match this period. The first constraint (PK) is the promise keeping constraint, stating that the value the worker gets from the contract either at the current job, through unemployment or at future jobs must deliver at least the promised value V_t. The second constraint (Budget) is the budget constraint of the worker, and the third constraint (BC) is the borrowing constraint. The last constraint (LC) is the worker and firm limited commitment constraint. We wrote the optimal contract taking as given the optimal search policy of workers, so the incentive compatibility constraint for search is implicit in the definition of $p(W,a)$ and $S(W,a)$.

Value of new matches Consider the value of a firm when it is just matched with a worker in market (v,a). The value from this match, denoted $\Pi_0(v,a_t)$, solves

$$\Pi_0(v,a_t) = \mathbb{E}_{\overline{x},x_0}\left[\max_{w_t,c_t,a_{t+1},V(x_{t+1})} x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi(V(x_{t+1}),a_{t+1},x_{t+1})|x_t\right]}{1+r}\right] \tag{3}$$

subject to

$$\text{(PK)}: \quad v \leq \mathbb{E}_{\overline{x},x_0}\left[u(c_t) + \beta\mathbb{E}_{x_{t+1}}\left[V_{t+1}(x_{t+1})|x_t\right]\right]$$

and the constraints (Budget) and (BC). All the choice variables are conditional on the realization of productivity $\overline{x}, x_0$. The firm fixed productivity is drawn from $\log\overline{x} \sim \mathcal{N}(0,\sigma_{\overline{x}}^2)$ and the initial value of firm productivity is drawn from $\log x_0 \sim \mathcal{N}(\log\overline{x},\sigma^2)$.

2.3 Equilibrium

Free entry Vacancy posting is subject to free entry. Firms have to pay a unit cost for posting a vacancy k so the free entry condition is

$$-k + \lambda_f(v,a)\Pi_0(v,a) \leq 0 \tag{4}$$

with equality for each active market (v,a).

Definition of an equilibrium An equilibrium is a set of value functions, policies and matching rates for each labor market (v,a) such that i) the unemployed worker policy maximizes the unemployment value, ii) the firm and employed worker policies satisfy the optimal contract, iii) the free entry condition is satisfied and iv) the job finding and vacancy filling rates are consistent with the matching function.

Denote the probability density function of the distribution of unemployed and employed workers by $\{D^u(a), D^e(V,a,x)\}$. In equilibrium, the laws of motion of distributions are satisfied given the policies.

3 The role of assets in wage contracts

I now characterize the optimal contract, with an emphasis on how assets alter the allocation relative to a model without assets (as in Burdett and Coles, 2003), and relative to the model studied in the unemployment insurance literature where assets play no role (as in Hopenhayn and Nicolini, 1997). I first derive a pseudo Euler equation, which shows that assets are used in the optimal contract to smooth the worker consumption and to influence the search decisions through wealth effects. I then show how, as a result, assets influence the degree of wage backloading in the optimal contract. Finally, I show that assets influence the optimal contracts only because neither firms nor workers can commit to transfers post EE and EU separations. In this section, I work under the assumption that assets are publicly observable, and I discuss this assumption in more length in section 5.

3.1 A pseudo Euler equation

I derive the optimality condition for assets, which shows that firms use them to smooth the consumption of workers and influence their search decision.

Combining the first-order condition for assets a_{t+1} and the envelope condition leads to an equation that depends on how the EE probability and the worker surplus from

EE transitions change with assets, that is $p_a(W_{t+1}, a_{t+1})$ and $S_a(W_{t+1}, a_{t+1})$. These terms reflect the fact that assets influence the search decision of workers $v(W_{t+1}, a_{t+1})$ and the matching rate in each market $\lambda_w(v_{t+1}, a_{t+1})$.

To further simplify this equation, we need to understand how assets influence job finding rates and worker mobility decisions. For this, we turn to the free entry condition that connects the value of new matches Π_0 and the vacancy filling rate λ_f (and thus the job finding rate λ_w through the matching function). Notice first that $\Pi_0(v, a_{t+1}, x_{t+1}) = \Pi_0(v, 0, x_{t+1}) + (1+r)a_{t+1}$, so the value of new matches increases in the asset of the worker. Notice next that the firm value Π_0 is strictly decreasing in the worker promised value v from the envelope condition. Thus, firms are indifferent between matching with a worker in a market with high value and high assets, and in a market with lower value but also lower assets. This indifference is captured by the following equation, derived from the free entry condition,

$$\partial_a \lambda_w(v_{t+1}, a_{t+1}) = -\partial_v \lambda_w(v_{t+1}, a_{t+1})(1+r)u'(c_{t+1}^{EE})$$

where c_{t+1}^{EE} is the consumption of the worker during the first period after an EE transition. This equation states that in equilibrium increasing the assets of workers by 1% has the same effect on the EE probability than inducing workers to search in a market where the worker value is $(1+r)u'(c_{t+1}^{EE})$% lower.

Now, combine this equation with the optimality condition for search v to get

$$S_a(W_{t+1}, a_{t+1}) = \partial_a \lambda_w(v_{t+1}, a_{t+1}) \left[v_{t+1} - W_{t+1}\right] = p(W_{t+1}, a_{t+1})(1+r)u'(c_{t+1}^{EE})$$

Intuitively, increasing assets by Δa makes workers more likely to match with another firm, and has a similar effect on the surplus from EE transitions than increasing consumption at the next job by $(1+r)\Delta a$. This equation shows why the optimality condition for assets a_{t+1} in the optimal contract can be rewritten as a *pseudo Euler equation*.

Proposition 1. *The optimal contract satisfies a pseudo Euler equation*

$$\frac{u'(c_t)}{\beta(1+r)} \geq \delta u(c_{t+1}^u) + (1-\delta)\mathbb{E}_{x_{t+1}}\left[p_{t+1}u'(c_{t+1}^{EE}) + (1-p_{t+1})u'(c_{t+1})|x_t\right] - \mathcal{W}_t \qquad (5)$$

where wealth effects on search are

$$\mathcal{W}_t \equiv \quad (1-\delta)\frac{u'(c_t)}{\beta(1+r)}\mathbb{E}_{x_{t+1}}\Bigg[\left(u'(c_{t+1})p_W(W_{t+1}, a_{t+1}) + \frac{p_a(W_{t+1}, a_{t+1})}{1+r}\right) \times \left(x_{t+1} - w_{t+1} + \frac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\Bigg]$$

Equation (5) holds with equality when the borrowing constraint does not bind.

Proof. See appendix A.2. □

This equation shows that assets are used in the optimal contract to smooth the consumption of workers, and to influence their search decision through wealth effects. In the next subsection, I show that the ability of firms to smooth the consumption of workers using assets dramatically changes the optimal contract.

3.2 Implications for tenure profiles

In the previous subsection I showed that firms use assets to smooth the consumption of workers. I now derive the implications for the path of wages, consumption and assets in the optimal contract.

The starting point is the following consumption growth condition

$$\frac{1}{u'(c_t)} - \frac{\beta(1+r)}{u'(c_{t-1})} = -\frac{p_W(W_t, a_t)}{1 - p(W_t, a_t)} \left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}} \left[\Pi(V(x_{t+1}), a_{t+1}, x_{t+1}) | x_t \right]}{1+r} \right) \quad (6)$$

which is derived in details in appendix A.1 from the optimality conditions of the contracting problem. This equation is similar to theorem 1 in Burdett and Coles (2003), lemma 3.2 in Shi (2009) and proposition 2 in Balke and Lamadon (2022), except that consumption c_t now replaces wages w_t on the left-hand side.

This condition equates the cost of backloading consumption on the left, with its benefits on the right. The cost is captured by the change in the inverse marginal utility of consumption, adjusted for the fact that firms and worker have different discount factors. When the utility function is more concave, the difference between $u'(c_t)$ and $u'(c_{t-1})$ is large and as a result the left-hand side is positive and large. The benefit of backloading consumption depends on the value that firms extract from matches, and on the derivative of the EE probability with respect to the worker value $p_W(W_t, a_t)$. By backloading the worker consumption, firms increase their continuation value W_t at the current job and induce workers to search in markets with higher values and lower job finding rate. Thus, backloading wages reduce the EE probability of workers. The incentives to backload wages are stronger when the firm continuation value is positive and large, and when increasing the worker value W leads to large changes in the EE probability (p_W is large).

When workers have no access to financial markets, $c_t = w_t$, firms face the following trade-off in designing the path of wages. On one hand, backloading wages helps to retain workers and thus increases expected firm profits. On the other hand, backloading wages

implies to backload consumption because workers have no access to savings. As a result, such a contract would seem very unattractive to workers with concave utility and firms would have to pay them higher average wages to be able to attract them. Therefore, the optimal degree of wage backloading balances the desire of firms to retain its workers, with the desire of workers for smooth consumption.

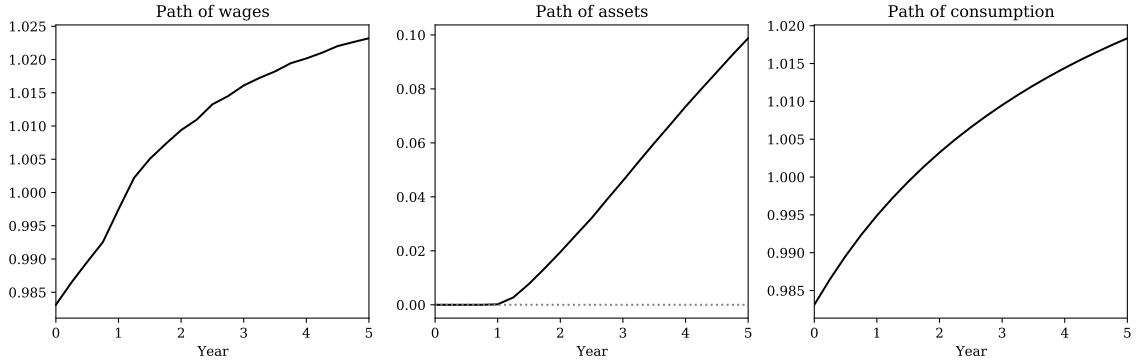
How does this trade-off change when workers have access to savings? Consider first how assets allow workers to smooth their consumption over time, as highlighted by equation (3.1). Firms know that workers can smooth their consumption even if their wages are backloaded by consuming their existing assets, or by borrowing if they did not face a borrowing constraint. Thus, a contract with backloaded wages does not seem as unattractive to workers anymore and firms have an additional incentive to backload wages relative to the model without assets. However, assets are also used to smooth consumption across states of the world, and in particular after an unemployment shock. Because firms cannot insure workers against such shocks, workers would like to accumulate precautionary savings to self-insure. Hence, it is costly for the worker to use her existing assets early during the match to smooth consumption over time, because this implies that her consumption will fall drastically if she were to become unemployed in the future. Instead, workers might even want to postpone their consumption to accumulate precautionary savings. Therefore, the optimal degree of wage backloading in the contract with assets features a trade-off between worker retention and precautionary savings. On one hand, firms want to backload wages to retain workers. On the other hand, backloading wages makes it hard for workers to accumulate precautionary savings, and thus makes contracts unattractive to risk-averse workers.

The pseudo Euler equation (3.1) also shows that firms use assets to influence the worker search decision through wealth effects. This is identical to Acemoglu and Shimer (1999) who describe how unemployment benefits and assets influence unemployed workers' search decision in a directed search model. For this reason, I describe this effect only briefly. In a directed search model, searching for a job is similar to selecting a lottery with payoff v and winning probability $\lambda_w(v, a)$. Whether workers select a lottery with high risk and high payoff or a lottery with low risk and low payoff depends on their preferences. With CRRA utility, workers with low assets are effectively more risk averse and thus select a safer lotteries in which jobs are easier to get but yield a lower value. How does this influence the contract? When the firm continuation value is positive, the firm wants to retain its worker and thus influences the search decision to reduce the EE rate. Therefore, relative to a model without wealth effects, the firm increases the asset of workers to make them select riskier lotteries, that is search in market with a higher value v

and lower job finding rate $\lambda_w(v, a)$. In section 4, I show that wealth effects on search are quantitatively minuscule so they play a minor role in shaping optimal contracts.

Figure 1 shows the optimal path of wages, assets and consumption for a newly hired worker from unemployment, starting with no assets $a_0 = 0$. In this specific example, the productivity of the firm happens to remain constant at $x_t = 1$ throughout. Wages increase continuously with tenure. Assets remain at the borrowing constraint for the first year of tenure, and then gradually increase over time. Finally, consumption increases over time but less than wages.

Figure 1: Tenure profiles of wages, assets and consumption



What explains the tenure profiles in figure 1? Initially, the continuation of firms is large so firms choose to backload wages significantly. If they could, they would even make workers borrow to smooth consumption. Instead, they choose to make workers consume their wages for the first year so workers are effectively hand-to-mouth at this point. The backloading of wages means that workers are initially hired with low wages but promised higher wages in the future, so the wage of workers rises over time. As a result, the continuation value of firms falls and it becomes less critical for firms to retain their workers. Besides, the gap between the wage of the worker and unemployment benefit b increases, which makes the precautionary savings stronger. As a result, it becomes optimal for firms to increase the assets of workers. This is achieved by a temporary increase in wage growth after one year of tenure. Meanwhile, consumption keeps increasing but not as fast as wages because workers now save some of their income.

3.3 Assets as a substitute for worker and firm commitment

The previous section showed that the optimal contract features a new trade-off between worker retention and precautionary savings when workers can trade risk-free bonds. I

now clarify the interaction between contracting frictions and assets by deriving conditions on the contracting environment under which assets influence the optimal allocation.

I first show that assets play no role in the optimal contract when firms can commit to transfers post EU transitions and when workers can commit to transfers post EE transitions.

Proposition 2. *Assume that firms and workers have commitment power so that τ_t^{ee} and τ_t^{eu} can be chosen freely in the optimal contract. Then, the path of consumption and EE probability $\{c_t, p_t\}$ are identical in optimal contract and in a restricted contract without assets, that is with the additional constraint $a_{t+1} = 0$.*

Proof. See appendix A.3. □

Proposition 2 clarifies why assets play a role in this model, whereas they usually play no role in the literature on optimal unemployment insurance when they are publicly observable by firms and workers. In this literature, at least since Hopenhayn and Nicolini (1997), it is common practice to assume that the government (the principal) can tax workers (the agent) when they find a job. This assumption is equivalent to assuming commitment on the side of workers. Besides, there is no exogenous separation between unemployment workers and the government in this literature (i.e. $\delta = 0$) so the transfers τ_t^{eu} play no role. Therefore, it is not surprising that assets play no role in the literature on unemployment insurance.

The next proposition clarifies the role that assets play when workers have limited commitment, as is always assumed in the context of wage contracts.

Proposition 3. *Assume that firms have commitment but workers do not ($\tau_t^{ee} = 0$), that productivity is constant within matches and that workers face no borrowing constraint. Then, the path of consumption and EE probability $\{c_t, p_t\}$ are identical in the contract with worker commitment, and in the contract with limited worker commitment and assets. Besides, the optimal path for assets satisfies*

$$a_{t+1} = -(\tau_{t+1}^{ee})^*$$

where $(\tau_t^{ee})^*$ denotes the optimal transfer in the model with worker commitment and no savings.

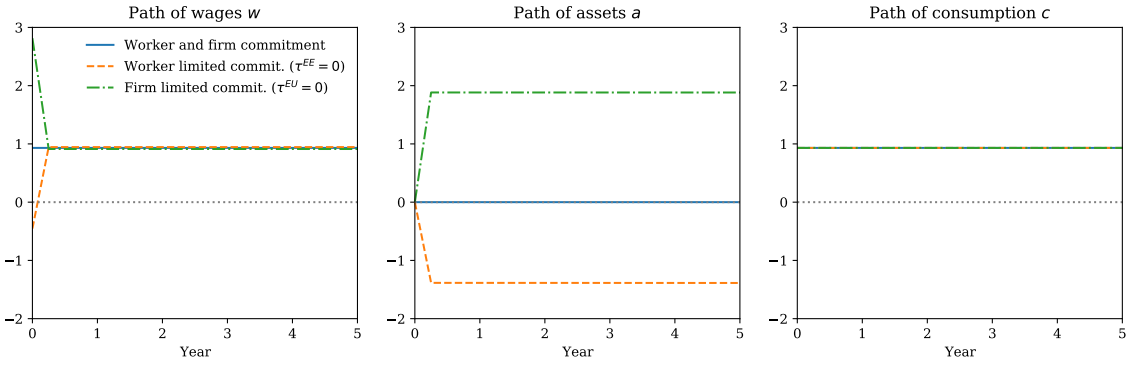
Proof. See appendix A.4. □

Proposition 3 shows that assets are a substitute for worker commitment and that the optimal contract achieves the same allocation with or without transfers post EE transitions. Thus, assets are a very powerful instrument to enforce optimal wage contracts despite the lack of worker commitment.

The allocation may be identical in these two contracts but the implementations are very different. To understanding why, remember that because of moral hazard, firms want to manipulate the continuation values of workers both at their current jobs and after an EE transition to influence their search decision. When workers have commitment, firms can do this using transfers from workers to firms after EE transitions. With limited commitment, transfers after EE transitions are no longer feasible but firms can instead influence the continuation value of workers at new jobs with assets. Specifically, by making the worker start their new jobs with less assets, firms reduce their wealth and therefore their future consumption. Proposition 3 shows that, in fact, we can achieve exactly the same allocation using either transfers or assets.

Figure 2 illustrates how the contract is implemented with limited worker commitment and assets. The dashed orange line shows the path of wages, assets and consumption with $\tau^{ee} = 0$. In this specific example, productivity is assumed to be constant within and across firms, firms and workers have the same discount rate, the worker is hired from unemployment with no initial asset and does not face any borrowing constraint. The optimal contract is implemented with a negative wage upon hiring, follows by a constant wage. Meanwhile, workers borrow to keep consumption constant[6]. Figure 2 also confirms that the allocation of consumption is the same with worker and firm commitment (the solid blue line) and with limited worker commitment and assets.

Figure 2: Equivalence between assets, worker and firm commitment



[6]In this specific example, introducing assets solves the model hazard because productivity is constant across firms. With heterogeneity in productivity, wages and consumption would not be constant over time even with worker commitment. This is because with constant wage the consumption of a worker matched with a low-productivity firm would jump up when she finds a better job. It is thus optimal for firms to insure the worker against this risk by increasing consumption at the current job, and reducing assets to reduce consumption at the future job. However, because the worker search decision is still private information, this brings back moral hazard. The optimal contract thus reduces the consumption and assets of workers over time, in a similar way as the government reduces unemployment benefits and increases taxes with unemployment duration in Hopenhayn and Nicolini (1997).

The next proposition shows that assets play a similar role when firms have limited commitment, that is they cannot make transfers to workers after unemployment shocks.

Proposition 4. *Assume that workers have commitment but firms do not ($\tau_t^{eu} = 0$). Then, the path of consumption and EE probability $\{c_t, p_t\}$ are identical in the contract with firm commitment, and in the contract with limited firm commitment and assets. Besides, the optimal path for assets satisfies*

$$a_{t+1} = (\tau_{t+1}^{eu})^*$$

where $(\tau_t^{eu})^*$ denotes the optimal transfer in the model with firm commitment and no savings, and the consumption after an unemployment shock c_t^u satisfies

$$(1+r)\beta u'(c_t^u) = u'(c_{t-1})$$

Proof. See appendix A.5. □

Proposition 4 shows that assets are also a substitute for firm commitment and that the optimal contract achieves the same allocation. It also shows that the consumption of workers does not fall when firms and workers discount the future at the same rate ($(1+r)\beta = 1$). With firm commitment, the contract is implemented with a positive transfer τ_t^{eu} from firms to workers after unemployment shocks. With assets, the contract is implemented using precautionary savings. The dotted green line in figure 2 shows the path of wages, assets and consumption in this contract without firm commitment. In this case, workers receive a large positive wage upon hiring, which allows them to accumulate enough precautionary savings right away to maintain consumption constant after an unemployment shock. An implication of this result is that introducing optimal wage contracts without worker mobility (or with worker commitment) in a model with precautionary savings unravels market incompleteness because firms undo the borrowing constraint of workers.

The model presented in section 3 has neither firm nor worker commitment so assets are used to substitute for both. As a result, wages are neither entirely backloaded as in proposition 3, nor entirely frontloaded as in proposition 4. Instead, they increase mildly with tenure, as in the data.

4 Quantitative implications

In this section, I estimate the model using matched employer-employee data from France. I then show that it is consistent with evidence that the wage growth of workers depends

on their wealth when they enter the labor market, and derive the implications for earnings inequality. I then use the model to quantify how much insurance workers receive inside and outside the firm against firm-level productivity shocks. Finally, I show that policies that improve the access of workers to financial markets, such as relaxing the borrowing constraint, crowd out the insurance that workers receive inside the firm.

4.1 Quantification

I quantify the model at quarterly frequency using matched employer-employee data from France between 2008 and 2019.

Data I combine annual data on firm balance sheet (FARE) with a panel of worker from social security data (DADS) containing 1/12th of the French labor force. I only keep in the sample workers with permanent full time contracts, and prime age workers (25-55 years old). I focus on private sector jobs in for-profit firms with at least 3 employees. The final sample contains approximately 530,000 workers and 130,000 firms per year.

Quantification strategy I quantify the model in two steps: first, I set some parameters externally; second, I infer the remaining model parameters by moment matching.

The model parameters set externally are the utility function, the foreign interest rate r and the matching function. The utility function is CRRA with coefficient $\gamma = 2$, following standard estimates from the macroeconomic literature. I set the interest rate to 4% annually. The matching function is Cobb-Douglas

$$\mathcal{M}(\phi_u + \kappa\phi_e, \phi_v) = B\left(\phi_e + \kappa\phi_u\right)^{\nu} \phi_v^{1-\nu}$$

with $\nu = 0.5$, which is an intermediate estimate between Menzio and Shi (2011) and Shimer (2005). $B = 0.26$ is calibrated to get a market tightness $\phi_v/(\phi_e + \kappa\phi_u)$ of 0.6, following Hagedorn and Manovskii (2008), given the job finding rate in my model[7].

The other model parameters are inferred by matching moments in the French data and in model-simulated data. Specifically, I simulate a panel of workers in the model and estimate the exact same set of moments in the model and in the data. The estimated parameters are the discount rate β, the vacancy posting cost k, the search efficiency on the job κ, the value of home production b, the exogenous separation rate δ, the persistence of productivity ρ_x, its volatility σ_x and the dispersion in productivity across matches $\sigma_{\bar{x}}$. These 8 parameters are estimated using 8 moments in the data.

[7] I do not have vacancy data for France and thus cannot estimate the matching function directly.

Moments	Data	Model	Parameters	
MPC from EU shock	30%	32%	Relative discount factor $\beta(1+r)$	0.995
Quarterly UE rate	20%	20%	Cost of posting vacancy k	0.55
Annual EE rate	6%	6.1%	On-the-job search efficiency κ	0.3
Annual EU rate	6%	6.1%	Separation rate δ	0.017
25-year cumulative wage growth	10%	10.9%	Flow unemployment value b	0.75
25-year cumulative ΔEE	-10%	- 8.6%	Dispersion in match productivity $\sigma_{\bar{x}}$	0.15
sd. of firm productivity growth	0.3	0.3	Volatility of productivity σ_x	0.14
Annual persistence of productivity	0.81	0.81	Persistence of productivity ρ_x	0.9

Table 1: Targeted moments in data vs. model and parameters

Table 1 shows the moments used in the estimation and the parameter values. I use moments on labor market flows (UE, EE and EU rates) to discipline the mobility of workers across jobs and the risk that workers face from unemployment. I use the tenure profile (25-year cumulative change) of log residual wages and EE transitions to discipline optimal contracts. I use moments on firm productivity, measured as value added per worker, to discipline the risk that firms face and might transmit to workers. Finally, I target an estimate of marginal propensity to consume of workers (MPC) after unemployment shocks of 30% quarterly, a standard value in the literature on precautionary savings, to discipline the access of workers to financial markets.

Relative to the literature on optimal wage contracts, the novel element in this model is worker access to financial markets. Thus, I now explain how the MPC is pinned down in the estimation. I follow an approach standard in the recent HANK literature (see Auclert et al., 2024), which consists in calibrating the discount rate of workers relative to the rate of return on assets $\beta(1+r)$ to match the MPC. When $\beta(1+r)=1$, workers accumulate an infinite amount of savings (as in Sotomayor, 1984, Chamberlain and Wilson, 2000) and as a result their MPC is null. When $\beta(1+r)$ is small, workers are relatively impatient and do not accumulate too much assets and as a result their MPC is positive. In the context of optimal contracts, this approach implies that promised value of workers will drift over time while employed. This can be seen from equation (6) when workers cannot switch jobs ($\kappa=0$), as the consumption of workers falls over time when $\beta(1+r)<1$ even in the absence of EE transitions.

Before turning to my main quantitative exercise, I use the model to confirm that wealth effects on search are negligible and to show that optimal wage contracts yield sizable benefits to firms relative to fixed-wage contracts.

Wealth effects on search The pseudo Euler equation 3.1 showed that firms manipulate assets to influence search through wealth effects. To assess whether this effect is large, I perform the following experiment. I first compute the path of wages shown in figure 1 for a worker hired from unemployment with no asset at a firm happens to remain constant at $x = 1$. I then compute the following one-shot deviation: every period, I let the worker change the search and savings decisions independently of firms assuming that the wage contract remains unchanged and that the worker will follow the recommended policy next period. The difference between the worker deviation and the optimal contract is due to the wealth effect on search $\mathcal{W}$ in equation 3.1. I find that the deviations are minuscule. For instance, over the first 20 years of tenure workers consume on average 0.007% more than is recommended by firms (at most 0.04%). This shows that wealth effects on search, while present, play a minor role in the optimal contract. This is the reason why I mostly abstracted from them in section 3. This finding is consistent with Shimer and Werning (2008) who also find that wealth effects on search are quantitatively small in the context of optimal unemployment insurance.

Comparison with fixed-wage contracts The existing literature on EE mobility sometimes assumes that firms use fixed-wage contracts for simplicity (e.g. in Burdett and Mortensen, 1998). Using the quantitative model, I evaluate whether this assumption is a strong restriction on the behavior of firms by computing the loss to a firm that adopts fixed-wage contracts relative to optimal contracts. Specifically, I take as given the equilibrium where all firms post optimal wage contracts. I then assume that a single firm deviates and posts contracts with a constant wage, where the wage level is such that a worker would be indifferent between working for this firm or for a firm offering the optimal contract. Finally, I compute the expected profits that this firm posting fixed-wage contracts would generate if it were matched with an unemployed worker with no initial asset.

I find that adopting a fixed-wage contract is equivalent to reducing productivity by 7% permanently, thus a significant loss to profitability. The reason for such a large difference in profits is that firms posting optimal wage contracts are able to retain workers more on average and especially when profits are high. For example, for a firm with the median level of fixed productivity $\bar{x} = 1$ the quarterly EE transition rate falls from 3.4 percent to 2.7 percent on average, and to 0.9 percent when profits are in the highest quintile because of productivity shocks.

4.2 Wealth at birth and life-cycle earnings inequality

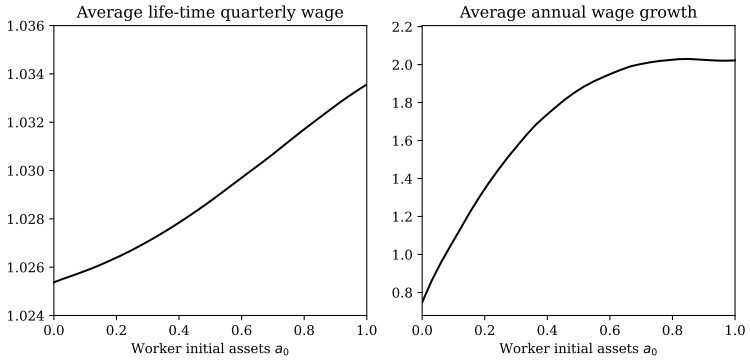
The model provides a new explanation for the widely documented persistence of income inequality across generations. For example, Solon (1992) shows that the rank of children's income is positively correlated with the income rank of their parents. The model generates this relation because the wage contract that workers receive depends on their initial assets when they enter the labor market. In particular, in the model wealthy workers choose to search in labor markets where wage contracts are more backlaoded because they can use their initial assets to smooth consumption. As a result, firms find it easier to retain these workers and, because of the free-entry condition, offer them higher average wages. By contrast, poor workers cannot afford to search for such high-paying jobs because they do not have the resources to smooth consumption early on in their career. Instead, they search for jobs where wages are less backloaded but also lower on average.

The relationship between initial worker assets, average wages and wage growth in the model is displayed in figure 3. As the worker assets increase from zero to about one quarter or average earnings, the average life-time earnings rises. At the same time, the average growth in annual wages over the first 5 years of the worker's career rises because wage contracts are increasingly backloaded. Some of these differences in life-cycle earnings across workers with different initial assets are attenuated by worker mobility across jobs. In particular, I find that the difference in total earnings between workers with no initial assets and workers with 2 quarters of income as initial assets is 0.04% in model simulations[8]. These differences rise to 0.20% when I prevent workers from switching jobs[9]. This happens because workers with little initial assets are more likely to switch jobs and thus climb the job ladder faster relative to wealthy workers.

[8]The differences are small in this version of the model because all jobs are identical ex-ante. The differences would be amplified if jobs have heterogeneous productivity ex-ante.

[9]In this counterfactual I keep the policy functions the same and I set the realization of EE transitions to 0 in all simulations.

Figure 3: Wealth at birth and life-cycle earnings inequality



The mechanism in the model is consistent with evidence on earnings growth across workers documented in the literature. In particular, Guiso et al. (2012) find, using Italian data, that firms that are financial constrained are more likely to backload the wages of managers and white-collar workers relative to blue-collar workers. They interpret this as evidence that firms take into account workers' ability to smooth consumption when they design wage contracts. Halvorsen et al. (2022) use administrative data from Norway and find that the children of parents with high net wealth experience higher wage growth than children of parents with low net wealth. An interpretation for this fact consistent with my model is that the wealth of parents allows workers to smooth consumption and search for jobs where wages grow faster because they are more backloaded, but also where wages are higher on average.

4.3 Inside vs. outside insurance against productivity shocks

I now quantify how much insurance workers receive inside and outside the firm against firm-level productivity shocks. I find that relative to a model with hand-to-mouth workers, the insurance that workers receive outside the firm crowds out significantly the insurance that they receive inside the firm.

Table 2 reports the pass-through of productivity shocks to consumption, wages and EE mobility and the corresponding marginal propensity to consumption (MPC) in the baseline model with assets and in a counterfactual with hand-to-mouth workers (HtM)[10]. This table shows the main empirical success of the model: it generates an MPC that is far less than 1 (22%) and a pass-through of productivity shocks to wages that is small (13.7%), both consistent with the data. By contrast, existing contracting models with HtM

[10]In the counterfactual model with HtM workers, all parameters are kept the same but we impose the constraint $a_{t+1} = 0$ on unemployed and employed workers.

	MPC	Pass-through to w	Pass-through to c	Pass-through to EE rate
Baseline with assets	22%	13.7%	3.0%	-7.6%
Counterfactual: HtM	100%	5.3%	5.3%	-3.4%

Table 2: Pass-through of productivity shocks to wages, consumption and EE mobility

workers imply an MPC of 1 and existing precautionary-savings models with exogenous contracts imply a pass-through of 1. It is critical for the model to generate a low MPC and a low pass-through of productivity shocks to wages because these numbers suggest that workers receive a lot of insurance against shocks from firms and on financial markets.

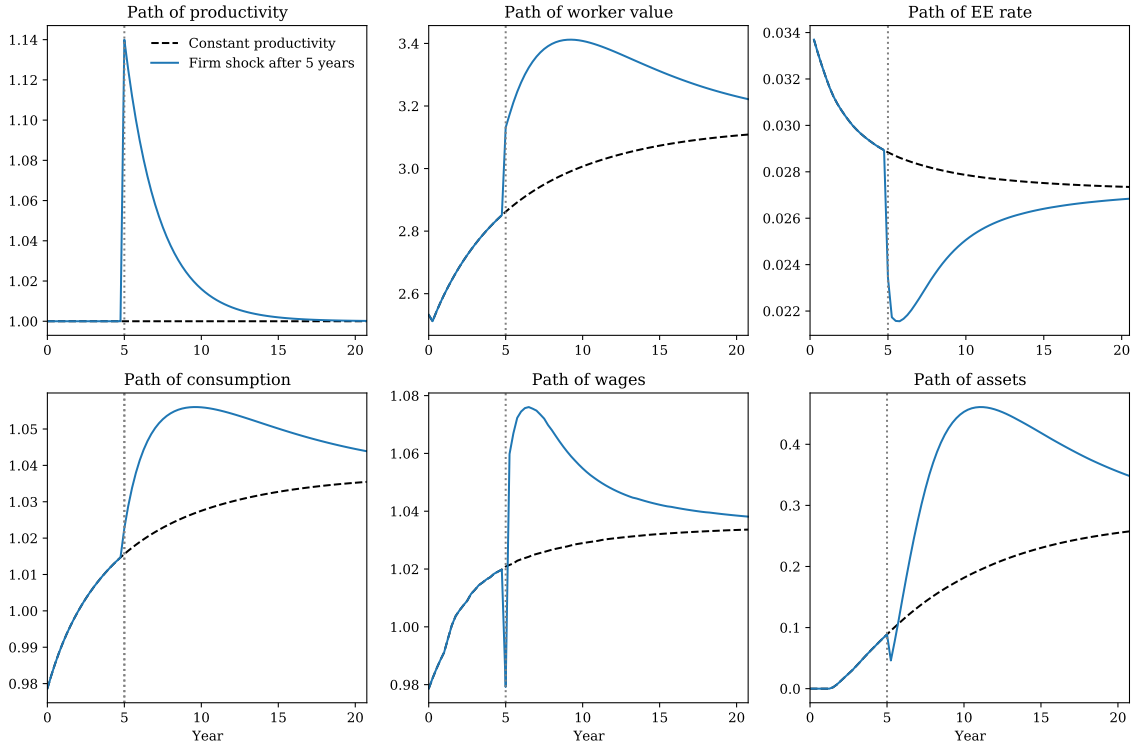
Table 2 also shows that the insurance that workers receive outside the firm significantly crowds out the insurance that they receive inside the firm. Specifically, in the baseline model the pass-through of productivity shocks to wages is 13.7% as opposed to 5.3% in the counterfactual model with HtM workers. However, workers receive more insurance overall in the baseline model since the pass-through to consumption is 3% as opposed to 5.3% in the counterfactual with HtM workers. The reason is that the MPC out of wage shocks is much lower in the baseline model (22% vs 100%). Thus, workers receive about 3 times (13.7/5.3) less insurance from firms when they can access financial markets but they receive about 2 times more insurance overall (3/5.3). Finally, table 2 shows that the worker EE transition rate falls by more after a positive productivity shock in the baseline model relative to the model with hand-to-mouth workers. As a result, firms are more likely to retain workers when they are more productivity and average match productivity is 0.5% higher in the baseline model relative to the counterfactual with HtM workers.

To understand these results, it is useful to consider the trade-off that firms face when they decide how to respond to firm-level productivity shocks. Remember that risk-neutral firms want to insure risk-averse workers against shocks by keeping consumption constant, and the best way to do this is to keep wages independent of productivity shocks. Insuring workers against risk allows firms to attract workers despite paying lower average wages. However, not responding to shocks means that the EE transition probability of workers would be independent of firm productivity. Thus, a worker would be just as likely to leave when productivity is high or when productivity is low. Instead, the firm could increase profits by paying workers relatively more when productivity is high to induce them to stay, and relatively less when productivity is low. Thus, the size of the pass-through is pinned down by this trade-off between insurance and optimizing worker retention. When workers have access to insurance outside the firm, they value stable wages less than when they are HtM because they can smooth consumption themselves.

As a result, firms choose to optimize worker retention relatively more and insure workers relatively less against risk. This is the reason why the pass-through to wages and to EE mobility are higher in the baseline model relative to the model with HtM workers.

To shed further light on the mechanism, I now describe the dynamics of the pass-through. Specifically, I compute the path of productivity, worker value, EE mobility rate, consumption, wage and assets of a worker hired from unemployment with no asset at a firm with fixed productivity $\overline{x} = 1$ and where productivity happens to remain constant over time. I then compute the same paths except that after 5 years of tenure, the firm productivity unexpectedly increases by σ_x, and then falls back to its initial level with persistence ρ_x. The results of this exercise are shown in figure 4, where the dotted black lines represent the paths with constant productivity, and the solid blue lines represent the paths with the productivity shock. The difference between the two lines can be interpreted as the pass-through of productivity shocks for that particular worker. The top row shows that in response to the shock, firms choose to increase the worker value at the current job in order to reduce the EE transition rate. The bottom row shows that the worker value increases because workers increase their consumption at the current job smoothly over time. This path of consumption is implemented as follows. Firms first cut the wage of workers on impact, and then promise workers much higher wages later on. Workers consume their existing assets initially to increase consumption despite lower wages, and then later use their higher wages to simultaneously increase their consumption and savings.

Figure 4: Dynamic pass-through of productivity shock



Perhaps the most surprising results here is that firms choose to *cut* wages on impact in response to a *positive* productivity shock. The reason is that firms choose to backload wages more when productivity increases. Remember that the degree of wage backloading is the result of a trade-off between worker retention and precautionary savings. When productivity increases, the worker retention motive becomes more important because firms generate more profits and do not want to lose their workers, whereas the precautionary savings motive remains the same. As a result, firms choose to backload wages more and reduce the worker assets despite exposing workers temporarily to more unemployment risk.

4.4 The crowding out of public insurance policies

In this section, I use the model to quantify the effects of a policy relaxing the borrowing constraint of workers, which aims at improving their ability to self-insure against shocks to their income and thus reduces the volatility of their consumption. I show that optimal wage contracts offset about half of the insurance gains for workers relative to a model where wage contracts are exogenous. Thus, improving the insurance that workers can receive outside the firm crowds out the insurance that workers receive inside the firm.

		MPC	Pass-through to w	Pass-through to c
New workers (experience $\leq$ 5 years)	Baseline $\underline{a} = 0$	37%	18.1%	6.7%
	Counterfactual $\underline{a} = -1$	25%	22.3%	5.5%
Experienced workers (experience > 5 years)	Baseline $\underline{a} = 0$	24%	14.6%	3.5%
	Counterfactual $\underline{a} = -1$	22%	15.3%	3.4%

Table 3: The effect of relaxing the borrowing constraint on wage volatility

I compare the baseline quantitative model where workers cannot borrow ($\underline{a} = 0$) to a counterfactual model where workers can borrow at rate r up to approximately 1 quarter of average wage income ($\underline{a} = -1$). I interpret this counterfactual as arising from a policy improving worker's access to financial markets, such as providing credit lines to workers[11]. In both models, I compute averages over simulated paths of 40 years for workers who start unemployed with 0 asset so as to capture the life-cycle dynamics of workers. It will be clear shortly why capturing these life-cycle dynamics is critical.

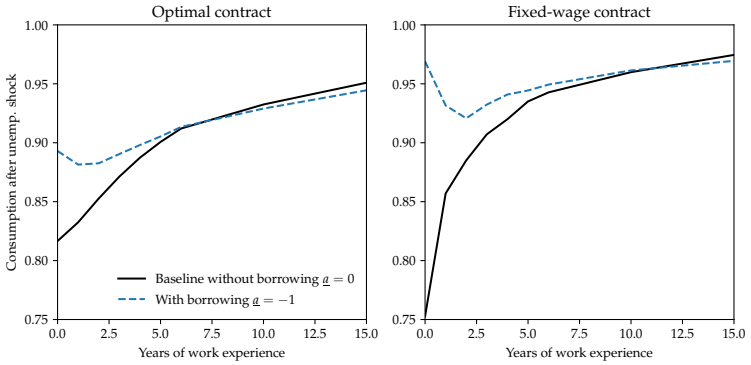
Table 3 reports the pass-through of productivity shocks to consumption and wages and the corresponding marginal propensity to consumption (MPC) in both models. The results are shown separately for new workers who just entered the labor force less than 5 years ago, and thus have not had much time to accumulate precautionary savings, and for more experienced workers who entered the labor force more than 5 years ago and less than 40 years ago. First, the results show that the MPC from wage shocks falls the most for new workers relative to experienced workers. Specifically, the MPC of new workers falls by a third, from 37% to 25%, whereas the MPC of experienced workers falls only from 24% to 22%. Thus, a policy that relaxes the borrowing constraint mostly benefits new workers who have not had the time to accumulate precautionary savings. Second, the results show that firms respond to the policy by passing productivity shocks through to wages more. In particular, for new workers the pass-through rises from 18.1% to 22.3%. As a result, the pass-through to consumption falls only from 6.7% to 5.5%. A back-of-the-envelope calculation shows that the pass-through to consumption would have fallen to $25\% \times 18.1\% = 4.5\%$ if the pass-through to wages had remained constant at 18.1% after the policy. Thus, the response of optimal contracts offsets about half of the insurance benefit of the policy ($(6.7\% - 5.5\%)/(6.7\% - 4.5\%) = 55\%$).

Figure 5 shows how the policy influences the insurance that workers receive against unemployment risk. The left panel reports the change in the consumption of workers

[11] In the counterfactual, the interest rate remains at r since this is a small open economy model. Solving for the interest rate r to generate the same level of assets does not alter the results.

after an unemployment shocks in the baseline model (black solid line) and the counterfactual (dashed blue line), as a function of the years of experience of workers. The right panel shows a similar exercise in a precautionary savings model with fixed-wage contracts similar to İmrohoroğlu (1989). In this precautionary savings model, workers transition between employment and unemployment with exogenous probability and they receive a constant wage when they are employed[12].

Figure 5: The effect of relaxing the borrowing constraint on unemployment risk



First, the results show that relaxing the borrowing constraint improves the insurance that workers receive against unemployment shocks because their consumption falls less when they become unemployed (the dashed blue line is higher than the solid black line). Here again, these effects are concentrated on new workers with less than 5 years of experience whereas more experienced workers do not benefit much from the policy because they have already accumulated precautionary savings and thus are further from the constraint. Second, the results show that optimal contracts offset about half of the gains of relaxing the borrowing constraint relative to a precautionary savings model where wage contracts are fixed. In particular, the average increase in consumption for new workers (difference between dashed blue line and solid black line) in the left panel is 4.4% whereas it is 9.3% in the right panel. The reason for this crowding out is that firms choose to backload wages more when workers can borrow because that helps them retain workers. As a result, workers find it harder to accumulate precautionary savings and their consumption falls by more when they become unemployed.

Taken together, these results suggest a significant amount of crowding out between the insurance that workers receive outside the firm, through financial markets, and the insurance that they receive inside the firm, through optimal wage contracts. For a policy

[12]The transition probabilities are calibrated to match the data and the discount factor of workers β is calibrated to match an average quarterly MPC of 30% against unemployment shocks, similar to the quantitative model with optimal contracts.

that improves the ability of workers to borrow, optimal contracts offset about half of the insurance gains to workers.

5 Hidden assets

Until now, I have assumed that the asset of workers is public information to firms. This assumption implies that firms know the assets that workers have when they match, and that firms and workers can contract on the saving decision of workers when they are employed. These implications are not very realistic, so it is natural to wonder whether the allocation would be very different if assets were private information to workers.

Two issues arise when assets are private information, which have been partially studied in the literature. First, workers might choose to direct their search in markets (v, a) even though their current asset is not a. This is an issue because firms expect workers to have assets a when they design wage contracts. This has been studied with fixed-wage contracts in Chaumont and Shi (2022) and Eeckhout and Sepahsalari (2023), but not with optimal contracts. Second, the saving decision of employed might differ from the recommended policy in the contract. This issue has been extensively studied in the optimal unemployment insurance literature (e.g. Werning, 2002, Abraham and Pavoni, 2008).

In this section, I characterize the deviations of workers when assets are private information. Specifically, I ask what deviation would a worker follow if she is offered the contracts designed under the assumption that assets are public information, while in fact the worker can lie about her existing assets when she matches with a firm and about her saving decision while employed. I first show that with CARA utility, there is no profitable deviation for workers so that the allocation is identical with hidden assets and publicly observable assets. I then show that workers benefit from deviation when utility if CRRA because of wealth effects on search. Throughout, I will emphasize results that differ from those already derived in the literature.

5.1 Equivalence with CARA utility: no wealth effect on search

I first show that with CARA utility $u(c) = -\gamma \exp(-\gamma c)$, the optimal contract when assets are public is also optimal when assets are private information to workers. This result is important because it shows that the assumption that assets are public information is not that critical for the analysis. In particular, we can interpret the model as one where firms propose a set of contracts (wages conditional on history of shocks and tenure) to workers, and where workers select the contracts they prefer depending on their assets and choose

to save and consume independently of firms.

Proposition 5. *Assume that utility is CARA and that workers face no borrowing constraint. Then, the equilibrium allocation with hidden assets is exactly identical than the allocation with publicly observable assets.*

Proof. See appendix A.6. □

This equivalence between private and public information arises because of the absence of wealth effect on search with CARA utility and it relies on two results. First, in the pseudo Euler equation 5 the terms capturing wealth effects is null, that is $\mathcal{W}_t = 0$. This result, extensively studied in the literature on optimal unemployment insurance, arises because two workers with the same wage contract but different levels of wealth are equally likely to switch job. Thus, the optimal contract with observable savings decision satisfies the Euler equation, and therefore it solves the optimal contract with hidden savings conditional on the initial asset of workers being known to firms.

Second, the absence of wealth effects means that workers have no incentive to report a different level of asset than their actual assets when they first match with firms. In particular, a worker with asset a never wants to under-report or over-report her assets by searching for a job in a market $(v, \tilde{a})$ where $\tilde{a} \neq a$. To see why, it is useful to assume that workers do not face any borrowing constraint first. In this case, it turns out that in any market (v_1, a_1) and (v_2, a_2) such that $v_1 \exp(\gamma r a_1) = v_2 \exp(\gamma r a_2)$, the job finding rate will be the same and workers will be offered the exact same contract. Thus, any deviation to a market $(\tilde{v}, \tilde{a})$ from (v, a) will yield the same value to workers than a deviation to a market $(\tilde{v} \exp(\gamma r(\tilde{a} - a)), a)$, i.e. a market with a different promised value but the true asset. But then, searching in such a market was already feasible for workers and was not optimal since (v) solves the search decision of workers in (2) conditional on their asset a. Therefore, workers would not benefit from deviating to a market where firms expect to meet workers with $\tilde{a}$ when their actual asset is a.

When workers face borrowing constraints, they too cannot benefit from reporting a level of asset $\tilde{a}$ different than their actual asset a. To see why, remember that firms take into account the worker's borrowing constraint when they design wage contract. When they expect it to bind, they do not backload wages as much because they know that workers cannot smooth consumption as much by consuming their existing assets. Thus, a worker with a lot of assets would receive a path of wages that is less backloaded by pretending to have very little assets. However, this worker would also receive lower average wages because firms generate lower expected profits in these markets. Overall, the worker is worse off. Conversely, a worker would not benefit by pretending to have more assets

than she actually have since in this case she would enjoy higher average wages but her consumption would be more backloaded. One way to interpret this result is by remembering that assets are used in the optimal contract as a substitute for the commitment power of workers. Ex-ante, workers would be better off if they could commit to transfers after EE transitions. Thus, they have no incentive to under-report their assets.

5.2 Optimal deviations with CRRA

I now show that the optimal contract when assets are public information is no longer incentive compatible when assets are private information and utility is CRRA. Specifically, I first solve for the optimal contract assuming that firms observe the worker's assets perfectly. I then ask how a worker would choose to deviate if she is offered this set of wage contracts. This exercise does not characterize the equilibrium with hidden assets but it helps assess whether the assumption that assets are public information is critical.

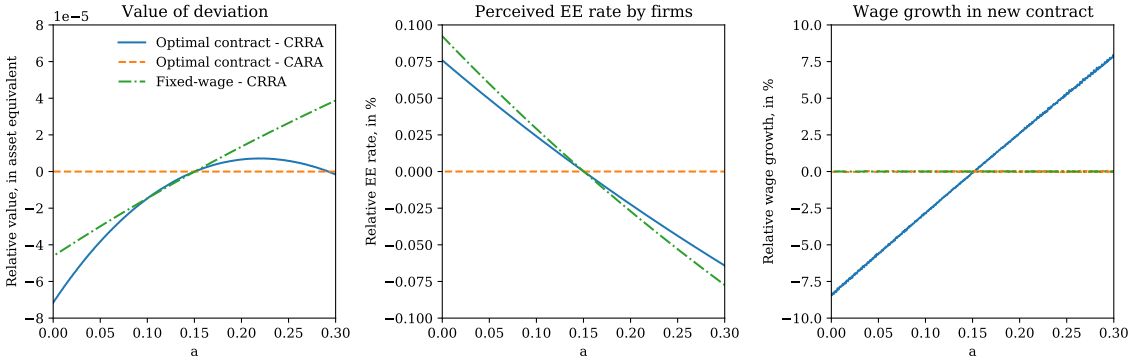
I first only briefly describe deviations in terms of savings that workers make while they are employed, since these have been extensively studied in the literature on optimal unemployment insurance. Equation (5) shows that when assets are public information, firms choose them to smooth the worker's consumption and to influence search through wealth effects $\mathcal{W}_t$. In particular, as explained in section 3.2, firms will make workers save more than the Euler equation would recommend when they want to retain them because this reduces the EE rate of workers. When the saving decision of workers is private information, the optimal deviation of workers is to follow their Euler equation, thus consume more today and save less. This deviation will make them more likely to switch jobs in the next period relative to what the firm would prefer.

I now describe the novel issue that arises in the context of wage contracts. Specifically, workers might deviate by searching in markets where the asset they are supposed to have $\tilde{a}$ is different than the asset they actually have a. The left panel of figure 6 illustrates this deviation by showing how much a worker with asset $a = 0.15$ would gain by searching in a market with $\tilde{a} \neq a$. For simplicity, the deviation is computed in a version of the model where workers only live for 2 periods, where they face no borrowing constraint and where productivity is constant across and within matches. I also compute contracts assuming that the saving decision is private information to stress the role that the initial asset of workers play[13]. The utility gains are measured in asset equivalent, that is how much additional assets would be needed to achieve the same gain than the deviation.

[13] I solve the optimal contract with hidden savings using the first-order approach, that is adding the Euler equation as a constraint on the optimal contract.

The blue line shows that workers benefit by searching in markets with higher assets, thus pretending to be wealthier. In this example, the optimal deviation for a worker with asset $a = 0.15$ is to search in markets indexed by $\tilde{a} = 0.22$. By contrast, the orange line confirms that workers do not benefit by deviating when utility is CARA. Finally, the green line shows the value of a deviation with CRRA utility but when firms are restricted to offer fixed-wage contracts, as in Chaumont and Shi (2022) and Eeckhout and Sepahsalari (2023). In this case, workers also benefit from searching in markets with higher assets $\tilde{a}$ but the value from deviations is monotonically increasing in $\tilde{a}$.

Figure 6: Optimal deviation with hidden assets



What accounts for these differences? The middle panel of figure 6 shows why workers benefits by deviating upward when utility is CRRA. It shows the EE rate of workers with different assets *a keeping the wage contract constant*, relative to the EE rate of workers with $a = 0.15$. By keeping the wage contract constant, we can evaluate how firms perceive the likelihood that they can retain workers as a function of the assets of workers. The figure confirms that workers are less likely to leave when their initial assets are high because of wealth effects on search: relatively rich workers search in markets with a relatively high value and low job finding rate. Because workers are perceived by firms as less likely to leave, the expected profit from matches rises. Because of the free entry condition, workers end up receiving higher average wages in these markets. This is the reason why workers benefit by searching in markets designed for workers with higher assets.

The right panel of figure 6 shows why the gains from deviating are not monotonic with optimal contract, whereas they are with fixed-wage contracts. Specifically, the right panel shows the growth rate of wages that workers receive in markets with different assets a. Because workers are perceived to be less risk-averse when they report higher initial assets a, firms optimally choose to backload wages more. This makes these contracts relatively unattractive to workers with lower assets because they would prefer consumption to be

smoother over time. As a result, workers choose to deviate upward but not too much. By contrast, when workers are offered fixed-wage contracts, they do not face this cost of deviating by assumption, and therefore the benefit from reporting higher assets is always increasing.

Thus, these results show that the optimal contract described in section 3 and 4 is not incentive compatible when assets are private information. However, deviations by workers are bounded when contracts are optimal and the gains from deviating appear to be very small in this 2-period example.

6 Conclusion

This paper studies the insurance that workers receive inside the firm, through wage contracts, and outside the firm, through financial markets, against firm-level productivity shocks and unemployment shocks. I build a new model with optimal contracts, assets, and search frictions in general equilibrium. In this model, firms take advantage of worker's access to financial markets when they design how wages change with tenure and after productivity shocks. When workers have access to insurance outside firms, firms backload wages more and provide less insurance to workers against productivity shocks. This generates a new source of earnings inequality across workers, and changes the implications of social insurance policies that improve workers' access to financial markets.

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Appendix

Contents

A Model appendix **A-1**

A.1 Consumption growth condition (6) A-1

A.2 Proof of proposition 1 . . . . . A-2

A.3 Proof of proposition 2 . . . . . A-3

A.4 Proof of proposition 3 . . . . . A-4

A.5 Proof of proposition 4 . . . . . A-6

A.6 Proof of proposition 5 . . . . . A-8

B Data appendix **A-8**

A Model appendix

A.1 Consumption growth condition (6)

Consider the optimal contract with $\tau_t^{ee} = \tau_t^{eu} = 0$ and denote the Lagrange multipliers on the constraints as $\eta_t, \lambda_t, \mu_t, \zeta_t$. The optimality conditions are

$$\begin{aligned}
w_t: &\quad (1-\delta)(1-p(W_t,a_t)) = \mu_t \\
c_t: &\quad \lambda_t u'(c_t) = \mu_t \\
V(x_{t+1}): &\quad (1-\delta)(1-p(W_t,a_t))(1+r)^{-1}\Pi_V(V(x_{t+1}),a_{t+1},x_{t+1}) + \beta\lambda_t = 0 \\
W_t: &\quad -p_W(W_t,a_t)\left(x_t - w_t + \tfrac{\mathbb{E}_{x_{t+1}}[\Pi(V(x_{t+1}),a_{t+1},x_{t+1})|x_t]}{1+r}\right) + \eta_t(1-p(W_t,a_t)) - \tfrac{\lambda_t}{1-\delta} = 0 \\
a_{t+1}: &\quad (1-\delta)(1-p(W_t,a_t))(1+r)^{-1}\mathbb{E}_{x_{t+1}}\left[\Pi_a(V(x_{t+1}),a_{t+1},x_{t+1})|x_t\right] = \mu_t - \zeta_t
\end{aligned}$$

and the envelope conditions are

$$\begin{aligned}
V_t: &\quad \Pi_V(V_t,a_t,x_t) = -\eta_t \\
a_t: &\quad \Pi_a(V_t,a_t,x_t) = -(1-\delta)p_a(W_t,a_t)\left(x_t - w_t + (1+r)^{-1}\mathbb{E}_{x_{t+1}}\left[\Pi(V(x_{t+1}),a_{t+1},x_{t+1})|x_t\right]\right) \\
&\qquad\qquad +\eta_t\left(\delta U'(a_t) + (1-\delta)S_a(W_t,a_t)\right) + \mu_t(1+r)
\end{aligned}$$

Consider the optimality conditions with respect to $V(x_{t+1})$ and W_t in the optimal contract, and the envelope condition with respect to V_t to get

$$\beta^{-1}(1+r)^{-1}\eta_{t+1} - \eta_t = -\frac{p_W(W_t,a_t)}{1-p(W_t,a_t)}\left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi(V(x_{t+1}),a_{t+1},x_{t+1})|x_t\right]}{1+r}\right)$$

Now combine the first-order conditions for c_t, w_t and $V(x_{t+1})$ and the envelope condition for V_t to get

$$\eta_{t+1} = \frac{\beta(1+r)}{u'(c_t)} \tag{A.1}$$

and therefore

$$\frac{1}{u'(c_t)} - \frac{\beta(1+r)}{u'(c_{t-1})} = -\frac{p_W(W_t,a_t)}{1-p(W_t,a_t)}\left(x_t - w_t + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi(V(x_{t+1}),a_{t+1},x_{t+1})|x_t\right]}{1+r}\right)$$

which is equation (6).

A.2 Proof of proposition 1

First, combine the first-order condition for a_{t+1} with the envelope condition for a_t to get

$$\begin{aligned} 1+r \geq \mathbb{E}_{x_{t+1}}\left[\eta_{t+1}\left(\delta U'(a_{t+1}) + (1-\delta)S_a(W_{t+1},a_{t+1})\right) + (1-\delta)(1-p(W_{t+1},a_{t+1}))(1+r)|x_t\right] \\ -\mathbb{E}_{x_{t+1}}\left[(1-\delta)p_a(W_{t+1},a_{t+1})\left(x_{t+1}-w_{t+1}+\frac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}),a_{t+2},x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\right] \end{aligned} \tag{A.2}$$

with equality if the borrowing constraint does not bind.

Next, combine the first-order condition for c_t and the envelope condition of the unemployed workers to get

$$U'(a_t) = (1+r)u'(c_t^u) \tag{A.3}$$

where c_t^u denotes the consumption of the unemployed worker.

Finally, we need to derive an expression for $S_a(W_{t+1},a_{t+1})$. From the envelope condition on the search problem of workers (2), we get

$$S_a(W_t,a_t) \equiv \kappa\left(v(W_t,a_t)-W_t\right)\partial_a\lambda_w(v(W_t,a_t),a_t)$$

We now derive an expression for $\partial_a\lambda_w(v(W_t,a_t),a_t)$ from the free entry condition of firms. Combining the first-order conditions and envelope conditions in the problem of new entrants (3) gives

$$\partial_v\Pi_0(v,a_t) = -\frac{1}{u'(c_t)}$$

$$\partial_a\Pi_0(v,a_t) = 1+r$$

Now consider the free entry condition

$$\lambda_f(v,a) = \frac{k}{\Pi_0(v,a)}$$

Differentiating this expression with respect to v and a gives

$$\partial_a\lambda_f(v,a) = -\lambda_f(v,a)\frac{1+r}{\Pi_0(v,a)}$$

$$\partial_v\lambda_f(v,a) = \lambda_f(v,a)\frac{1}{\Pi_0(v,a)u'(c_t)}$$

Taking the ratio gives

$$\partial_a\lambda_f(v,a) = -\partial_v\lambda_f(v,a)(1+r)u'(c_t)$$

With a constant returns to scale matching function, we can express the job finding rate as $\lambda_w(v,a) = f(\lambda_f(v,a'))$. Therefore,

$$\partial_a\lambda_w(v,a) = -\partial_v\lambda_w(v,a)(1+r)u'(c_t)$$

We can use this expression to rewrite $S_a(W_t,a_t)$ as

$$S_a(W_t,a_t) = -(1+r)u'(c_t^{EE})\kappa\left(v(W_t,a_t)-W_t\right)\partial_v\lambda_w(v(W_t,a_t),a_t)$$

where c_t^{EE} is the consumption of the worker during the first period after an EE transition. We can further simplify this term using the first-order condition of the search problem

$$\lambda_w(v_t, a_t) + \partial_v \lambda_w(v_t, a_t)\,(v_t - W_t) = 0$$

and get

$$S_a(W_t, a_t) = (1+r)u'(c_t^{EE})\kappa\lambda_w(v(W_t, a_t), a_t) = (1+r)u'(c_t^{EE})p(W_t, a_t) \tag{A.4}$$

Combine equations (A.2), (A.1), (A.3), and (A.4), to get

$$\begin{aligned} u'(c_t) \geq\ & \mathbb{E}_{x_{t+1}}\left[\beta(1+r)\left(\delta u'(c_{t+1}^u) + (1-\delta)p(W_{t+1}, a_{t+1})u'(c_{t+1}^{EE})\right) + (1-\delta)(1-p(W_{t+1}, a_{t+1}))u'(c_t)|x_t\right] \\ & -(1-\delta)u'(c_t)\mathbb{E}_{x_{t+1}}\left[\frac{p_a(W_{t+1}, a_{t+1})}{1+r}\left(x_{t+1} - w_{t+1} + \frac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\right] \end{aligned}$$

For the final step, rewrite the consumption growth condition (6) evaluated at $t+1$ as

$$u'(c_t) = \beta(1+r)u'(c_{t+1}) - u'(c_t)u'(c_{t+1})\frac{p_W(W_{t+1}, a_t)}{1-p(W_{t+1}, a_t)}\left(x_{t+1} - w_{t+1} + \frac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right)$$

We can replace $u'(c_t)$ on the right-hand side by this expression and get

$$\begin{aligned} u'(c_t) \geq\ & \beta(1+r)\left(\delta u'(c_{t+1}^u) + (1-\delta)\mathbb{E}_{x_{t+1}}\left[p(W_{t+1}, a_{t+1})u'(c_{t+1}^{EE}) + (1-p(W_{t+1}, a_{t+1}))u'(c_{t+1})|x_t\right]\right) \\ & -(1-\delta)u'(c_t)\mathbb{E}_{x_{t+1}}\left[\left(u'(c_{t+1})p_W(W_{t+1}, a_t) + \frac{p_a(W_{t+1}, a_{t+1})}{1+r}\right)\left(x_{t+1} - w_{t+1} + \frac{\mathbb{E}_{x_{t+2}}[\Pi(V(x_{t+2}), a_{t+2}, x_{t+2})|x_{t+1}]}{1+r}\right)|x_t\right] \end{aligned}$$

which is equation (5). QED.

A.3 Proof of proposition 2

Consider any allocation that solves the optimal contract with a path for assets $a_{t+1}(x^t)$, where x^t denote the history of productivity since the beginning of the match. We can construct a new path wages $\tilde{w}_t(x^t)$ and transfers $\tilde{\tau}_t^{ee}(x^t)$, $\tilde{\tau}_t^{eu}(x^t)$ such that the allocations $c_t(x^t)$, $p_t(x^t)$ remain the same and solve the optimal contract while the path for assets satisfies $\tilde{a}_{t+1}(x^t) = 0$.

Let the new paths of transfers be $\tilde{\tau}_t^{ee}(x^t) = \tau_t^{ee}(x^t) - a_t(x^{t-1})$ and $\tilde{\tau}_t^{eu}(x^t) = \tau_t^{eu}(x^t) + a_t(x^{t-1})$. The new path of wages is $\tilde{w}_t(x^t) = (1+r)a_t(x^{t-1}) + w_t(x^t) - a_{t+1}(x^t)$ for all periods except the first, and $\tilde{w}_t(x_t) = w_t(x_t) - a_{t+1}(x_t)$ for the first period.

First, we show that the worker value remains constant after any history with this new contract. The worker value satisfies

$$\begin{aligned} \tilde{V}_t(x^t) =\ & \delta U(\tilde{\tau}_t^{eu}(x^t)) + (1-\delta)\left[\tilde{W}_t(x^t) + S(\tilde{W}_t(x^t), -\tilde{\tau}_t^{ee}(x^t))\right] \\ =\ & \delta U(a_t(x^{t-1}) + \tau_t^{eu}(x^t)) + (1-\delta)\left[\tilde{W}_t(x^t) + S(\tilde{W}_t(x^t), a_t(x^{t-1}) - \tau_t^{ee}(x^t))\right] \end{aligned}$$

with $\tilde{W}_t(x^t) = u(c_t(x^t)) + \beta\mathbb{E}_{x_{t+1}}\left[\tilde{V}_{t+1}(x^{t+1})|x^t\right]$. This shows that $\tilde{V}_t(x^t) = V_t(x^t)$ and $\tilde{W}_t(x^t) = W_t(x^t)$ after any x^t.

Then, we show that this new contract achieves the same initial value to firms. Note that the firm value will differ after some histories. The firm value after history x^t under the original contract satisfies

$$
\begin{aligned}
\Pi_t(x^t) = \quad & (1-\delta)\left(1-p(W_t(x^t), a_t(x^{t-1}) - \tau_t^{ee}(x^t))\right)\left(x_t - w_t(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right) \\
& -\delta(1+r)\tau_t^{eu}(x^t) + (1-\delta)p(W_t(x^t), a_t(x^{t-1}) - \tau_t^{ee}(x^t))(1+r)\tau_t^{ee}(x^t) \\
= \quad & (1-\delta)\left(1-p(W_t(x^t), -\tilde{\tau}_t^{ee}(x^t))\right)\left(x_t - \tilde{w}_t(x^t) + (1+r)a_t(x^{t-1}) - a_{t+1}(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right) \\
& -\delta(1+r)\left(\tilde{\tau}_t^{eu}(x^t) - a_t(x^{t-1})\right) + (1-\delta)p(W_t(x^t), -\tilde{\tau}_t^{ee}(x^t))(1+r)\left(\tilde{\tau}_t^{ee}(x^t) + a_t(x^{t-1})\right)
\end{aligned}
$$

Now define $\tilde{\Pi}_t(x^t) = \Pi_t(x^t) - (1+r)a_t(x^{t-1})$ for all x^t. Use $\tilde{W}_t(x^t) = W_t(x^t)$ in the previous equation to get

$$
\begin{aligned}
\tilde{\Pi}_t(x^t) = \quad & \Pi_t(x^t) - (1+r)a_t(x^{t-1}) \\
= \quad & (1-\delta)\left(1-p(\tilde{W}_t(x^t), -\tilde{\tau}_t^{ee}(x^t))\right)\left(x_t - \tilde{w}_t(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right) \\
& -\delta(1+r)\tilde{\tau}_t^{eu}(x^t) + (1-\delta)p(\tilde{W}_t(x^t), -\tilde{\tau}_t^{ee}(x^t))(1+r)\tilde{\tau}_t^{ee}(x^t)
\end{aligned}
$$

which shows that $\tilde{\Pi}_t(x^t)$ is precisely the firm value under the alternative contract. Finally, we can check that the value of new entrants is the same under the alternative and the original contracts

Now plug this in the problem of new entrants to verify that the firm value is the same in the first period

$$
\begin{aligned}
\tilde{\Pi}_0(v, a_t) = \quad & \mathbb{E}_{x_t}\left[x_t - \tilde{w}_t(x_t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x^t\right]}{1+r}\right] \\
= \quad & \mathbb{E}_{x_t}\left[x_t - w_t(x_t) + a_{t+1}(x_t) - a_{t+1}(x_t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right] \\
= \quad & \mathbb{E}_{x_t}\left[x_t - w_t(x_t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right] \\
= \quad & \Pi_0(v, a_t)
\end{aligned}
$$

Therefore, the new contract with $a_{t+1}(x^t) = 0$ after any x^t achieves the same initial firm value. The allocations satisfies the constraints on the optimal contract every period, including the borrowing constraint, so the new contract is also a solution to the optimal contract. Finally, the EE transition rate $p_t(x^t)$ remains the same since

$$
p(\tilde{W}_t(x^t), -\tilde{\tau}_t^{ee}(x^t)) = p(W_t(x^t), a_t(x^{t-1}) - \tau_t^{ee}(x^t))
$$

This shows that the optimal contract can be solved with the constraint $a_{t+1}(x^t) = 0$ without loss of generality. QED.

A.4 Proof of proposition 3

The proof requires productivity to remain constant within matches so I will denote the wage as $w_t(x)$ in a slight abuse of notation, where x denotes the fixed productivity of the match and t denotes time. Denote the allocation under worker (and firm) commitment as $c_t(x), p_t(x)$ achieved with some path of transfers $\tau_t^{ee}(x), \tau_t^{eu}(x)$. We assume that $a_{t+1}(x) = 0$, which is without loss of generality from proposition 2.

We now construct a new contract consisting of paths for wages, assets and transfers $\tilde{w}_t(x), \tilde{a}_{t+1}(x), \tilde{\tau}_t^{ee}(x), \tilde{\tau}_t^{eu}(x$ such that

a) $\tilde{\tau}_t^{ee}(x) = 0$ for all t,

b) the worker consumption $c_t(x)$ and values $V_t(x), W_t(x)$ are the same for all t,

c) the worker EE transition rate $p_t(x) \equiv p(W_t(x), a_t(x) - \tau_t^{ee}(x))$ is the same for all t,

d) the initial firm value $\Pi_0(v, a)$ is the same.

This new contract delivers the same value to firms and satisfy all the constraints of the contract with worker commitment. Thus, it solves the relaxed problem with worker commitment. Since it also satisfy the additional constraint $\tau_t^{ee} = 0$, it must also solve the optimal contract with limited worker commitment. Thus, the optimal contract with assets and limited worker commitment delivers the same path of consumption and EE rate than the optimal contract with worker commitment.

Construct the new contract $\tilde{w}_t(x), \tilde{a}_{t+1}(x), \tilde{\tau}_t^{ee}(x), \tilde{\tau}_t^{eu}(x)$ using

$$\tilde{w}_t(x) = w_t(x) - (1+r)\tilde{a}_t(x) + \tilde{a}_{t+1}(x)$$

$$\tilde{a}_{t+1}(x) = -\tau_{t+1}^{ee}(x)$$

$$\tilde{\tau}_t^{ee}(x) = 0$$

$$\tilde{\tau}_t^{eu}(x) = \tau_t^{eu}(x) - \tilde{a}_t(x)$$

except for the first period where $\tilde{w}_t(x) = w_t(x) + \tilde{a}_{t+1}(x)$. The consumption of the worker implied by this path is unchanged since

$$\tilde{c}_t(x) = (1+r)\tilde{a}_t(x) + \tilde{w}_t(x) - \tilde{a}_{t+1}(x) = w_t(x) = c_t(x)$$

for all periods except the first, and

$$\tilde{c}_t(x) = (1+r)a_0 + \tilde{w}_t(x) - \tilde{a}_{t+1}(x) = (1+r)a_0 + w_t(x) = c_t(x)$$

for the first period.

We now show that the worker value remains constant after any history with this new contract. The worker value satisfies

$$\begin{aligned}\tilde{V}_t(x) = &\ \delta U(\tilde{a}_t(x) + \tilde{\tau}_t^{eu}(x)) + (1-\delta)\left[\tilde{W}_t(x) + S(\tilde{W}_t(x), \tilde{a}_t(x))\right] \\ = &\ \delta U(\tau_t^{eu}(x)) + (1-\delta)\left[\tilde{W}_t(x) + S(\tilde{W}_t(x), -\tau_t^{ee}(x))\right]\end{aligned}$$

with $\tilde{W}_t(x) = u(c_t(x)) + \beta\tilde{V}_{t+1}(x)$. This shows that $\tilde{V}_t(x) = V_t(x)$ and $\tilde{W}_t(x) = W_t(x)$ for all t. It also follows that the EE rate $p_t(x)$ remains the same since

$$p_t(x) \equiv p(\tilde{W}_t(x), \tilde{a}_t(x)) = p(W_t(x), -\tau_t^{ee}(x))$$

Finally, consider the firm value with these paths, denoted $\tilde{\Pi}_t(x)$. It satisfies

$$\begin{aligned}\tilde{\Pi}_t(x) = &\ (1-\delta)\left(1 - p(\tilde{W}_t(x), \tilde{a}_t(x))\right)\left(x - \tilde{w}_t(x) + \tfrac{\tilde{\Pi}_{t+1}(x)}{1+r}\right) - \delta(1+r)\tilde{\tau}_t^{eu}(x) \\ = &\ (1-\delta)\left(1 - p(W_t(x), -\tau_t^{ee}(x))\right)\left(x - w_t(x) + (1+r)\tilde{a}_t(x) - \tilde{a}_{t+1}(x) + \tfrac{\tilde{\Pi}_{t+1}(x)}{1+r}\right) - \delta(1+r)\tau_t^{eu}(x) + \delta(1 \\ = &\ (1-\delta)\left(1 - p(W_t(x), -\tau_t^{ee}(x))\right)\left(x - w_t(x) - \tilde{a}_{t+1}(x) + \tfrac{\tilde{\Pi}_{t+1}(x)}{1+r}\right) - \delta(1+r)\tau_t^{eu}(x) \\ &\ +(1+r)\tilde{a}_t(x) + (1-\delta)p(W_t(x), -\tau_t^{ee}(x))(1+r)\tau_t^{ee}(x)\end{aligned}$$

Now guess and verify that $\tilde{\Pi}_t(x) = \Pi_t(x) + (1+r)\tilde{a}_t(x)$. Going back to the first period, we get

$$\begin{aligned}
\tilde{\Pi}_0(v, a_t) &= \mathbb{E}_x\left[x - \tilde{w}_t(x) + \tfrac{\tilde{\Pi}_{t+1}(x)}{1+r}\right] \\
&= \mathbb{E}_x\left[x - w_t(x) - a_{t+1}(x) + a_{t+1}(x) + \tfrac{\Pi_{t+1}(x)}{1+r}\right] \\
&= \mathbb{E}_x\left[x - w_t(x) + \tfrac{\Pi_{t+1}(x)}{1+r}\right] \\
&= \Pi_0(v, a_t)
\end{aligned}$$

and therefore the initial value of the firm is the same with this new contract. QED.

Why the equivalence breaks down with borrowing constraints or productivity shocks In the example of figure 2, the optimal contract with limited worker commitment makes the worker borrow to fund a negative wage payments at $t = 0$. With borrowing constraints, this allocation is no longer feasible and the contract cannot implement the allocation with worker commitment. Thus, the equivalence breaks down.

When productivity varies during the match, the equivalence breaks down because with limited commitment, assets are non-contingent on the realization of productivity whereas with worker commitment, transfers are contingent on the realization of productivity. Thus, transfers are a more flexible instrument than assets.

A.5 Proof of proposition 4

Denote the allocation under firm commitment as $c_t(x^t), p_t(x^t)$ achieved with some path of transfers $\tau_t^{ee}(x^t), \tau_t^{eu}(x^t)$, where x^t denotes the history of shocks since the beginning of the match. Without loss of generality, assume that $a_{t+1}(x^t) = 0$. Denote also $c_t^u(x^t)$ the consumption of workers after an unemployment shock.

We now construct a new contract consisting of paths for wages, assets and transfers $\tilde{w}_t(x^t), \tilde{a}_{t+1}(x^t), \tilde{\tau}_t^{ee}(x^t), \tilde{\tau}_t^{eu}$ such that

a) $\tilde{\tau}_t^{eu}(x^t) = 0$ for all x^t,

b) the worker consumption $c_t(x^t)$ and values $V_t(x^t), W_t(x^t)$ are the same for all x^t,

c) the worker EE transition rate $p_t(x^t) \equiv p(W_t(x^t), a_t(x^{t-1}) - \tau_t^{ee}(x^t))$ is the same for all x^t,

d) the worker consumption after unemployment shock $c_t^u(x^t)$ is the same,

e) the initial firm value $\Pi_0(v, a)$ is the same.

This new contract delivers the same value to firms and satisfy all the constraints of the contract with firm commitment. Thus, it solves the relaxed problem with firm commitment. Since it also satisfy the additional constraint $\tau_t^{eu} = 0$, it must also solve the optimal contract with limited firm commitment. Thus, the optimal contract with assets and limited firm commitment delivers the same path of consumption and EE rate than the optimal contract with firm commitment.

Construct the new contract $\tilde{w}_t(x^t), \tilde{a}_{t+1}(x^t), \tilde{\tau}_t^{ee}(x^t), \tilde{\tau}_t^{eu}(x^t)$ using

$$\tilde{w}_t(x^t) = w_t(x^t) - (1+r)\tilde{a}_t(x^t) + \tilde{a}_{t+1}(x^t)$$

$$\tilde{a}_{t+1}(x^t) = \tau_{t+1}^{eu}(x^{t+1})$$

$$\tilde{\tau}_t^{ee}(x^t) = \tau_t^{ee}(x^t) + \tilde{a}_t(x^{t-1})$$

$$\tilde{\tau}_t^{eu}(x^t) = 0$$

except for the first period where $\tilde{w}_t(x^t) = w_t(x^t) + \tilde{a}_{t+1}(x^t)$. Note that we can allow for productivity shocks here because severance payments are independent of the realization of productivity shocks at $t+1$, that is $\tau_{t+1}^{eu}(x^{t+1}) = \tau_{t+1}^{eu}(x^t)$. Thus, we can set $\tilde{a}_{t+1}(x^t) = \tau_{t+1}^{eu}(x^{t+1})$ in the new contract. As before, the consumption of the worker implied by this new contract is unchanged.

We now show that the worker value remains constant after any history with this new contract. The worker value satisfies

$$\begin{aligned}\tilde{V}_t(x^t) = &\quad \delta U(\tilde{a}_t(x^{t-1}) + \tilde{\tau}_t^{eu}(x^t)) + (1-\delta)\left[\tilde{W}_t(x^t) + S(\tilde{W}_t(x^t), \tilde{a}_t(x^{t-1}) - \tilde{\tau}_t^{ee}(x^t))\right] \\ = &\quad \delta U(\tau_t^{eu}(x^t)) + (1-\delta)\left[\tilde{W}_t(x^t) + S(\tilde{W}_t(x^t), -\tau_t^{ee}(x^t))\right]\end{aligned}$$

with $\tilde{W}_t(x^t) = u(c_t(x^t)) + \beta \mathbb{E}_{x_{t+1}}\left[\tilde{V}_{t+1}(x^{t+1})|x^t\right]$. This shows that $\tilde{V}_t(x^t) = V_t(x^t)$ and $\tilde{W}_t(x^t) = W_t(x^t)$ for all x^t. It also follows that the EE rate $p_t(x^t)$ remains the same since

$$p_t(x^t) \equiv p(\tilde{W}_t(x^t), \tilde{a}_t(x^{t-1}) - \tilde{\tau}_t^{ee}(x^t)) = p(W_t(x^t), -\tau_t^{ee}(x^t))$$

Next, we show that the consumption after unemployment shock is the same. This follows directly from the envelope condition of the unemployed worker

$$U'(\tilde{a}_t(x^{t-1})) = U'(\tau_t^{eu}(x^t)) = (1+r)u'(c_t^u(x^t)) \tag{A.5}$$

Finally, consider the firm value with these paths, denoted $\tilde{\Pi}_t(x^t)$. It satisfies

$$\begin{aligned}\tilde{\Pi}_t(x^t) = &\quad (1-\delta)\left(1 - p(\tilde{W}_t(x^t), \tilde{a}_t(x^{t-1}) - \tilde{\tau}_t^{ee}(x^t))\right)\left(x_t - \tilde{w}_t(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x^t\right]}{1+r}\right) \\ &\quad -\delta(1+r)\tilde{\tau}_t^{eu}(x^t) + (1-\delta)p(\tilde{W}_t(x^t), \tilde{a}_t(x^{t-1}) - \tilde{\tau}_t^{ee}(x^t)))(1+r)\tilde{\tau}_t^{ee}(x^t) \\ = &\quad (1-\delta)\left(1 - p(W_t(x^t), -\tau_t^{ee}(x^t))\right)\left(x_t - w_t(x^t) + (1+r)\tilde{a}_t(x^{t-1}) - \tilde{a}_{t+1}(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x^t\right]}{1+r}\right) \\ &\quad +(1-\delta)p(W_t(x^t), -\tau_t^{ee}(x^t))(1+r)(\tau_t^{ee}(x^t) + \tilde{a}_t(x^{t-1})) \\ = &\quad (1-\delta)\left(1 - p(W_t(x^t), -\tau_t^{ee}(x^t))\right)\left(x_t - w_t(x^t) - \tilde{a}_{t+1}(x^t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x^t\right]}{1+r}\right) \\ &\quad +(1+r)\tilde{a}_t(x^{t-1}) - \delta(1+r)\tau_t^{eu}(x^t) + (1-\delta)p(W_t(x^t), -\tau_t^{ee}(x^t))(1+r)\tau_t^{ee}(x^t)\end{aligned}$$

Now guess and verify that $\tilde{\Pi}_t(x^t) = \Pi_t(x^t) + (1+r)\tilde{a}_t(x^{t-1})$. Going back to the first period, we get

$$\begin{aligned}\tilde{\Pi}_0(v, a_t) = &\quad \mathbb{E}_{x_t}\left[x_t - \tilde{w}_t(x_t) + \frac{\mathbb{E}_{x_{t+1}}\left[\tilde{\Pi}_{t+1}(x^{t+1})|x^t\right]}{1+r}\right] \\ = &\quad \mathbb{E}_{x_t}\left[x_t - w_t(x_t) - a_{t+1}(x_t) + a_{t+1}(x_t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right] \\ = &\quad \mathbb{E}_{x_t}\left[x_t - w_t(x_t) + \frac{\mathbb{E}_{x_{t+1}}\left[\Pi_{t+1}(x^{t+1})|x^t\right]}{1+r}\right] \\ = &\quad \Pi_0(v, a_t)\end{aligned}$$

and therefore the initial value of the firm is the same with this new contract.

Finally, we compute consumption growth after unemployment shocks. Combining the envelope condition (A.5), equation (A.1) and the optimality condition with respect to τ_t^{eu} in the optimal contract gives

$$(1+r)\beta u'(c_t^u) = u'(c_{t-1})$$

QED.

A.6 Proof of proposition 5

The proof has two steps. First, we show that conditional on an initial value for worker assets a, the optimal allocation of the contract when assets are public information also solves the optimal contract when assets are private information. Second, we show that workers do not benefit by searching in markets indexed by assets $\tilde{a}$ when their actual asset is a. Together, these two results show that the optimal contract with public information is also optimal if assets were private information.

Details coming soon.

B Data appendix

I use administrative data provided by the CASD in France between 2008 and 2019. My analysis relies on two main files:

a) the panel version of the "DADS tous salariés" database, containing detailed information about employment history for 1/12th of the French population every year;

b) "FARE" database, with annual information about firm balance sheet and income statement for the entire private sector except firms in the agricultural sector

I complement my analysis with information about the structure of firms ("Contours des entreprises profilées") provided by the CASD and with national account information on depreciation rates and the price index provided by INSEE.

Sample selection From the FARE file on firms, I exclude firms with invalid information (e.g. missing ID), firms belonging to the public sector and household employers. I also drop firms from the financial sector because it is particularly challenging to estimate productivity for these firms as their income is mostly reported in their financial statement, unlike other firms. One challenge with this data is that it is reported at the legal unit level ("UL"), and several legal units can belong to the same firm. Since I want to measure EE transitions across firms competing for the same workers, it is important that I aggregate firms within coherent economic units. To do so, I use information from the "Entreprise profilée" ("EP") files for available years, and extrapolate the information back in time when necessary.

From the DADS file, I exclude interns and apprenticeships as well as workers from the public sectors or working for non-profits. I keep prime-age workers (25 to 55 years old) and workers with full-time positions and permanent contracts (CDI). I focus on relatively stable jobs because I study the problem of worker retention, and it would not fit very well the case of temporary contracts (CDD) since they usually end after a short period of time. In my sample I find that full-time workers with permanent contracts account for about 60% of private sector jobs.

I merge the worker and firm data together and find that 95% of workers are successfully matched to a firm. I restrict my sample to workers and firms who at in the panel for at least 3 years and for firms with at least 3 employees (in the panel or not). I drop firms with negative or missing labor productivity and those with labor productivity growth below and above the 0.5 and 99.5 percentiles respectively. I also drop individuals with wage growth below or above the 0.5 and 99.5 percentiles.

Definition of labor productivity I measure labor productivity as value added per worker, adjusted for the cost of capital

$$\text{LP} = \frac{\text{sales + variation in shocks - cost of materials - cost of capital}}{\text{number of employees}}$$

Sales includes products, services and merchandises sold while the number of employees is the average full-time equivalent number of workers in that year. The data contains information about depreciation costs reported by firms, but this information is known to be sensitive to accounting strategies followed by firms. Instead, I construct my own estimates for the cost of capital as follows. I first measure the depreciation rate at the year-industry level using national accounts data on consumption and stock of fixed capital (average of 6.5% annual). I then add the average interest rate paid by firms on their debt in my dataset for firms with positive debt (average of 10%) and multiply with firm tangible assets reported in the firm data.

I residualize the log productivity on dummies for firm-age to control for a life-cycle component. My measure of labor productivity is closely related to the accounting measure of operating profits, and therefore not surprisingly their correlation is very strong both across firms and over time within firms.

I decompose labor productivity into an aggregate, a sectoral and a firm component by assuming that they are log-additive

$$\log y_{jst} = \log a_t + \log z_{st} + \log x_{jst} \tag{A.6}$$

I measure aggregate productivity $\log a_t$ by average across firms each year. I then measure sectoral productivity $\log z_{st}$ by averaging the residual across firms within sector each year. Finally, firm-level productivity $\log x_{jst}$ is estimated as the residual. I confirm visually that there are no trends in sectoral productivity.

Definition of wages I define wages as daily labor earnings using the worker total worker earnings net of payroll taxes but gross of income taxes. This includes regular wages, overtime pay, bonuses and even payment in kind. It excludes however stock options, but these are less omnipresent in France than they are in the U.S. Note also that medical insurance is not a major component of pay in France, unlike in the U.S.

I divide total labor earnings in a year by the number of days worked at that firm. The data contains information about hours but for workers with full-time jobs and permanent contracts it usually refers to the legal number of hours and therefore does not represent the actual number of hours worked. For this reason I do not adjust for it.

Definition of labor market flows Identifying EE transitions is challenging because workers sometimes hold multiple jobs at the same time. For this reason, I first identify the main job of a worker defined as the job with the earliest start date. I drop jobs that lasted for less than 35 hours during a year (a regular work week) and main jobs if they end up accounting for less than 50% of total earnings from simultaneous jobs. I also drop individuals with more than 5 jobs in a given year.

I use the exact start and end dates of jobs to identify a job transition. An EE transition occurs if the new job starts 18 days or less after the previous job ends. This leaves a little bit of room for workers who take 2 weeks of holidays in between jobs. The risk is that it might also include workers who transit through unemployment for just 2 weeks and find a new job quickly. Note

however that France is a country in which the job finding rate is fairly low (I estimate 20% per quarter) so most likely this risk is minimal. I also count as EE transitions if the new and old jobs overlap for some time (i.e. the worker holds 2 jobs for some time), but my results are robust to remove them from the sample.

An important moment that I target in my quantitative exercise is the share of EE transitions with positive wage growth. This moment is important because it is informative about why workers change jobs, and therefore has important implications for the retention elasticity. In France it is common for workers to change jobs to receive severance payments and compensations for vacations not taken when they switch job. As a result, average daily earnings at the current job is often larger than average daily earnings at the next job because it includes these extraordinary payments on top of the wage. Indeed, I compute that only 40% of workers experience a positive wage growth when daily earnings are computed in this naive way, and I find that workers who are about to make an EE transition experience an average wage growth of 8%, compared to 1% for the entire population. To control for these exceptional payments, I compute the share of job transitions with a positive wage change by comparing daily labor earnings at the new job with daily labor earnings at the previous job the previous year. I use the same method in the model.

When a worker separates from their previous jobs and does not make an EE transition, I define it as a separation into non-employment. When a worker from my sample moves to another job that is not in my sample (e.g. transition from private sector to public sector), I do not count it either as an EE transition nor as a separation into non-employment nor as a stayer.

I compute the duration of non-employment as the number of months until a worker reappears in my sample, conditional on the worker reappearing. By conditioning on whether a worker ever comes back in my sample I sort out workers who leave the labor force permanently (e.g. retirement, death). I only estimate this moment on the first half of my sample (2008-2015) so that workers have plenty of time to come back.